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Jackson et al.

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(54) **WEARABLE INTERACTIVE AUDIO DEVICE**

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(51) **Int. Cl.**
H04R 1/02 (2006.01)
G06F 3/01 (2006.01)

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CPC **H04R 1/02** (2013.01); **G06F 3/016** (2013.01); **G06F 3/017** (2013.01); **G06F 3/165** (2013.01); **H04R 1/1016** (2013.01); **H04R 3/007** (2013.01)

(58) **Field of Classification Search**

CPC H04R 1/02; H04R 1/1016; H04R 2225/61; H04R 3/007

See application file for complete search history.

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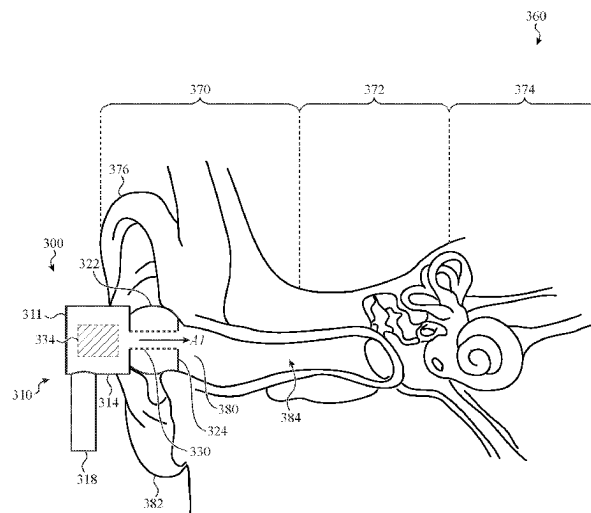
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(57) **ABSTRACT**

Embodiments are directed to a wearable audio device, such as an earbud. The earbud may be configured to detect input using various sensors and structures. For example, the earbud may be configured to detect gestures, physical manipulations, and so forth performed along or on the earbud. In response to the detected inputs, the earbud may be configured to change various outputs, such as an audio output or a haptic output of the device. The earbud may also include a microphone to register voice commands. In some cases, the microphone may be used to control the earbud using the registered voice command in response to one or more detected gestures or physical manipulations.

20 Claims, 19 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/055,071, filed on Aug. 4, 2018, now Pat. No. 10,757,491.

- (60) Provisional application No. 62/683,594, filed on Jun. 11, 2018.

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H04R 3/00 (2006.01)

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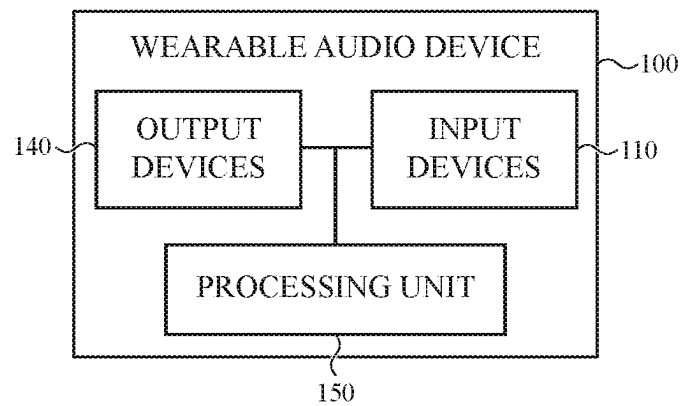


FIG. 1A

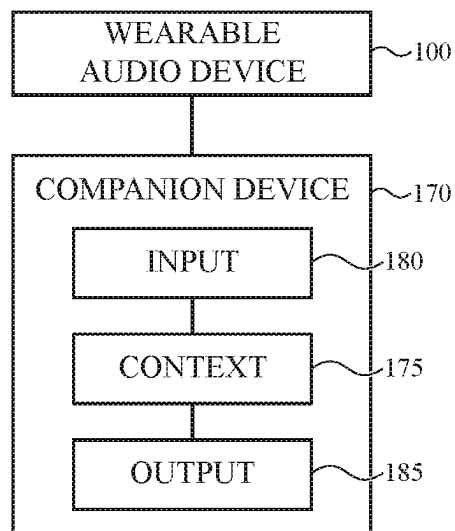
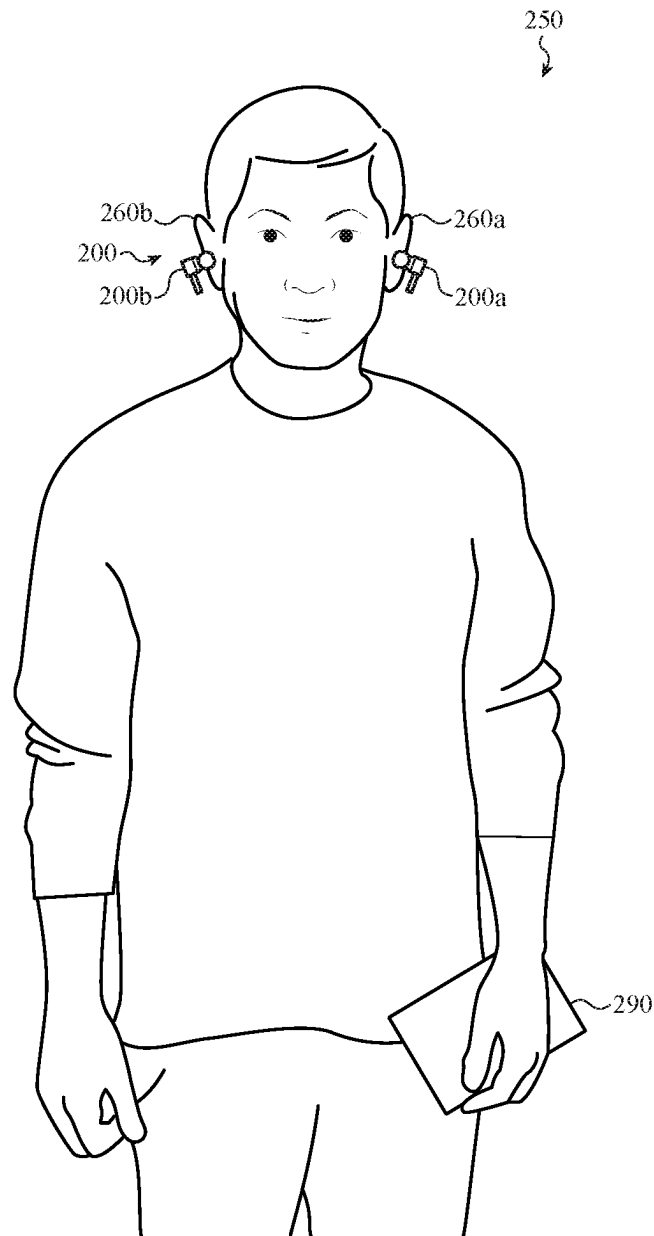


FIG. 1B

**FIG. 2**

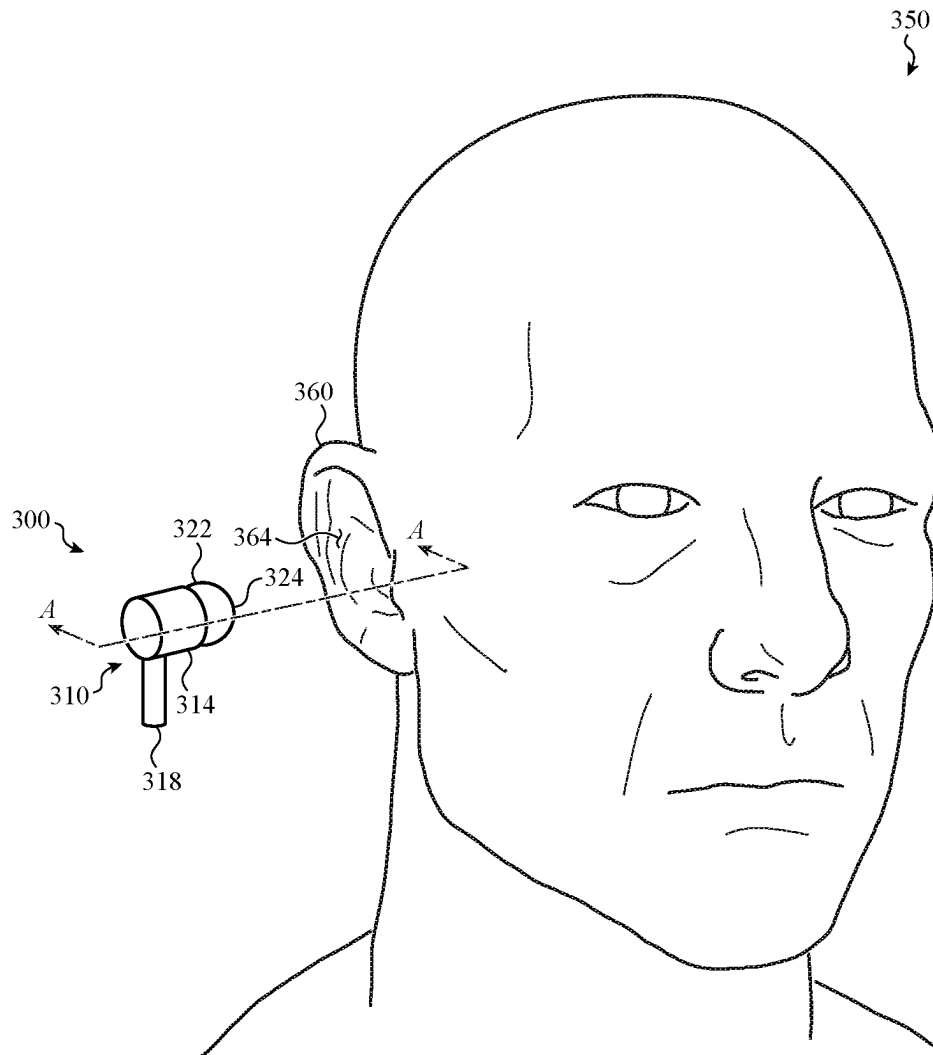


FIG. 3A

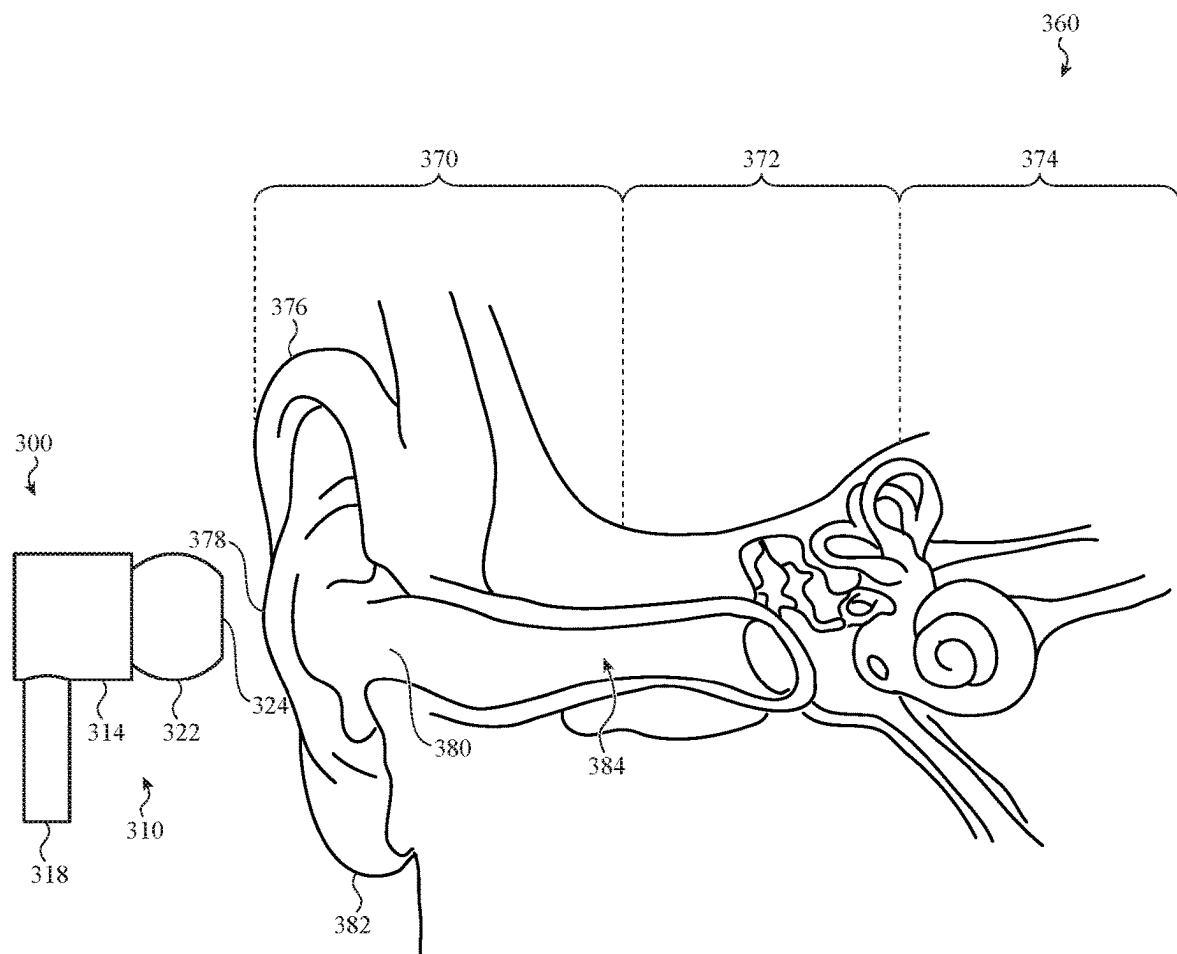


FIG. 3B

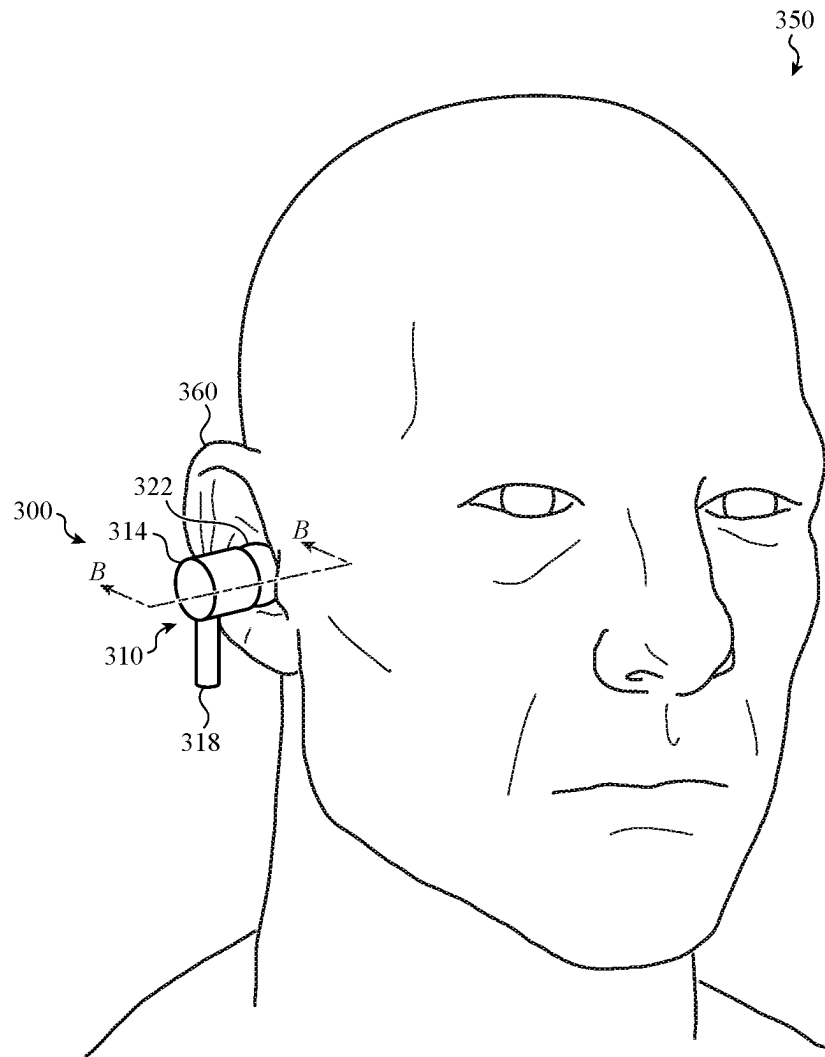


FIG. 4A

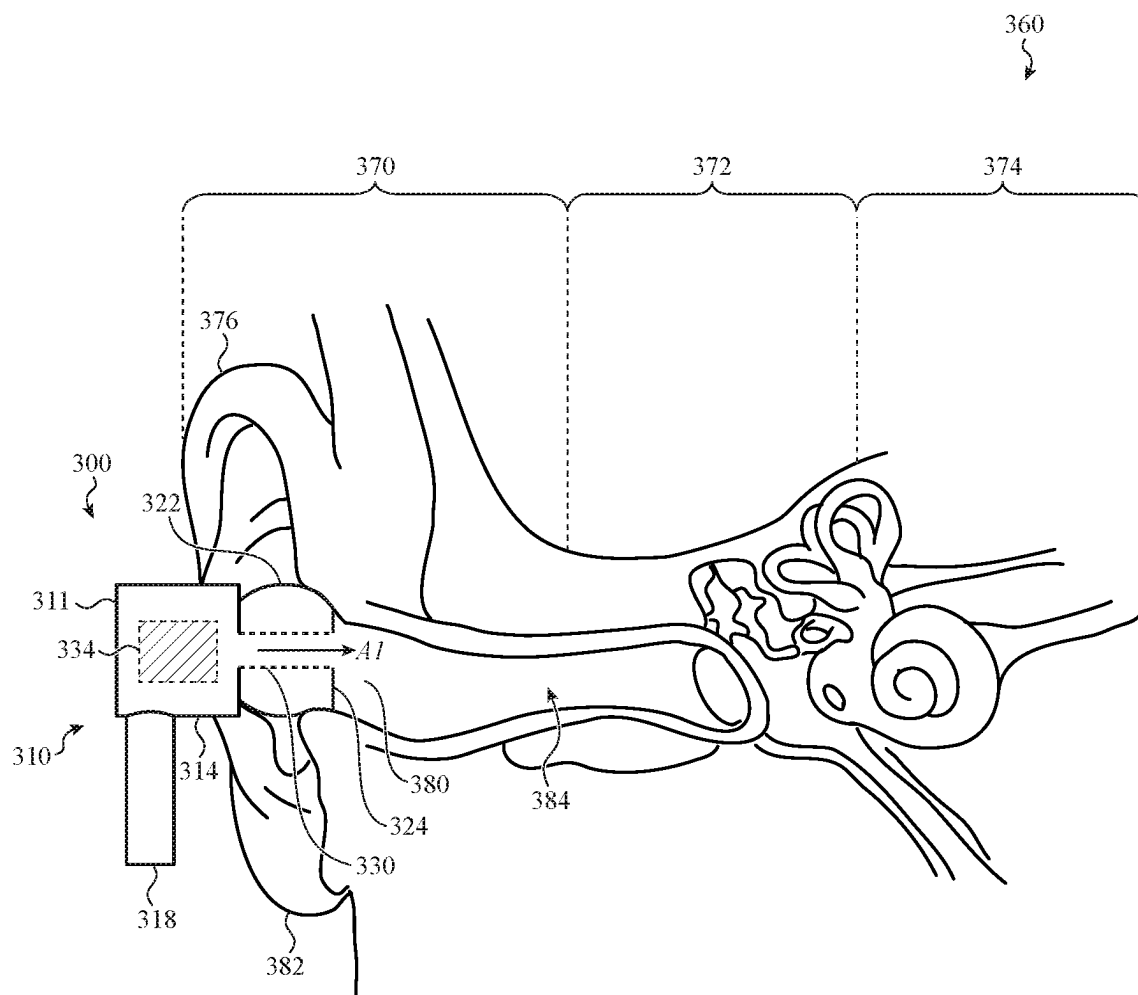


FIG. 4B

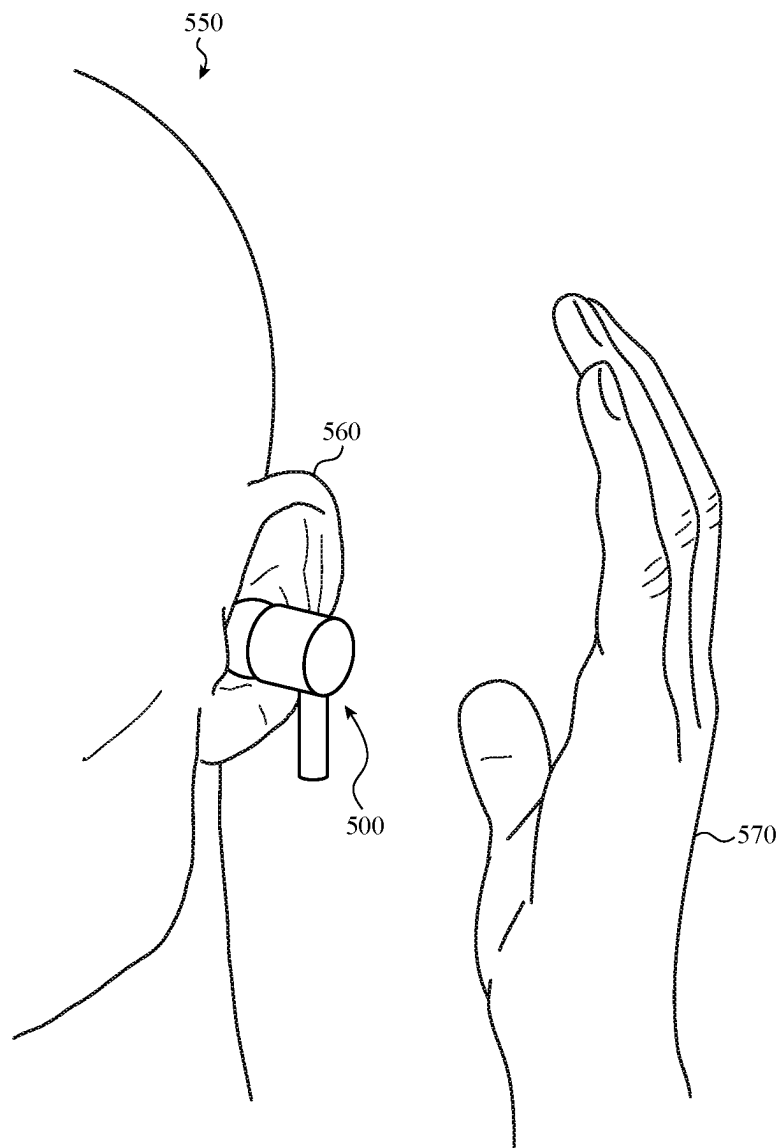


FIG. 5

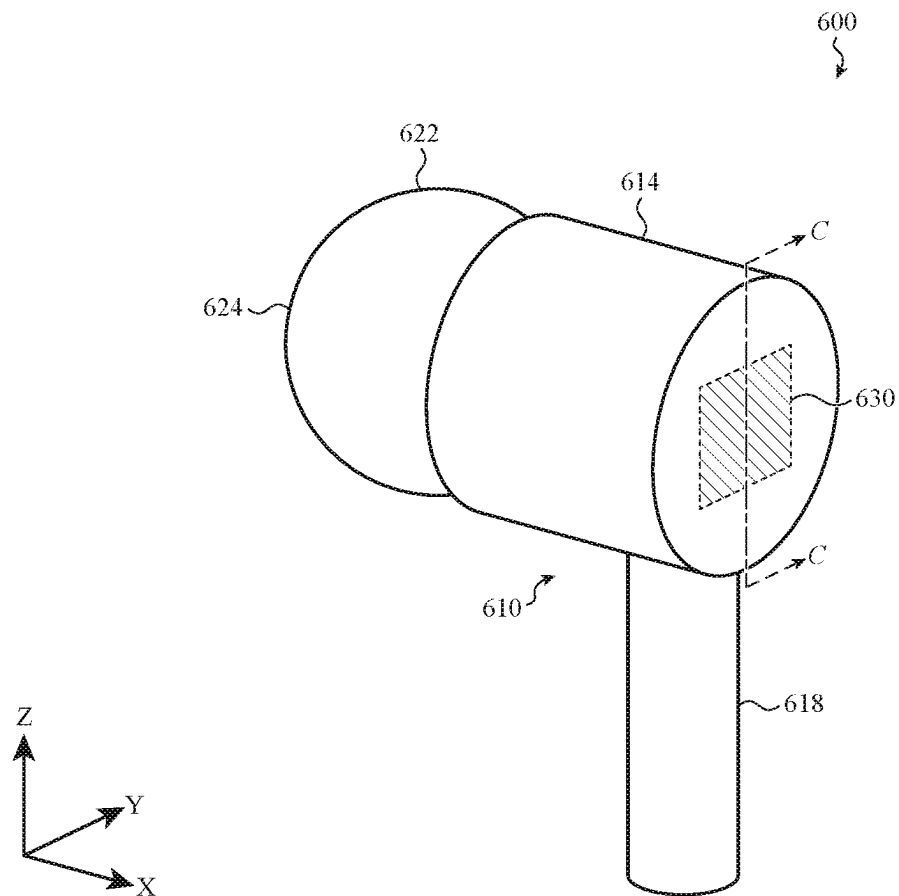


FIG. 6A

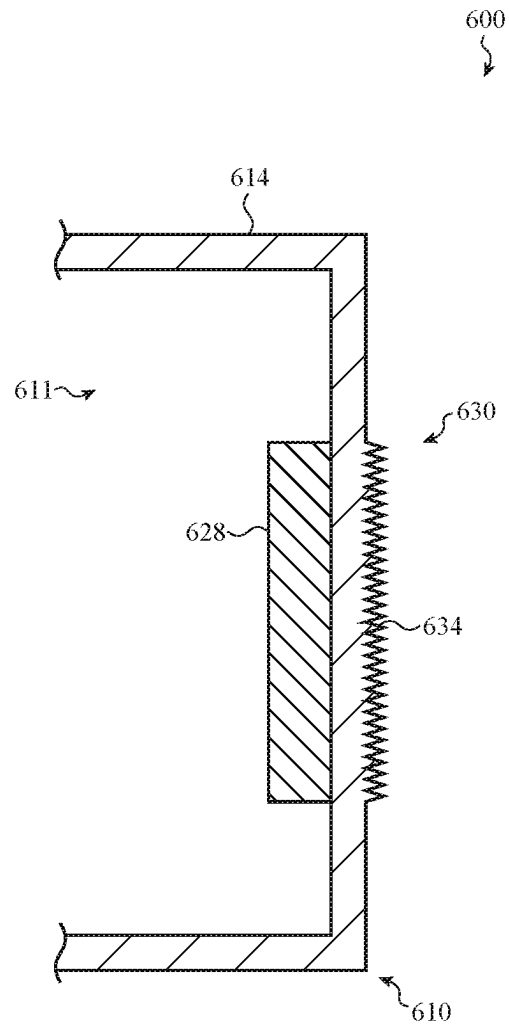


FIG. 6B

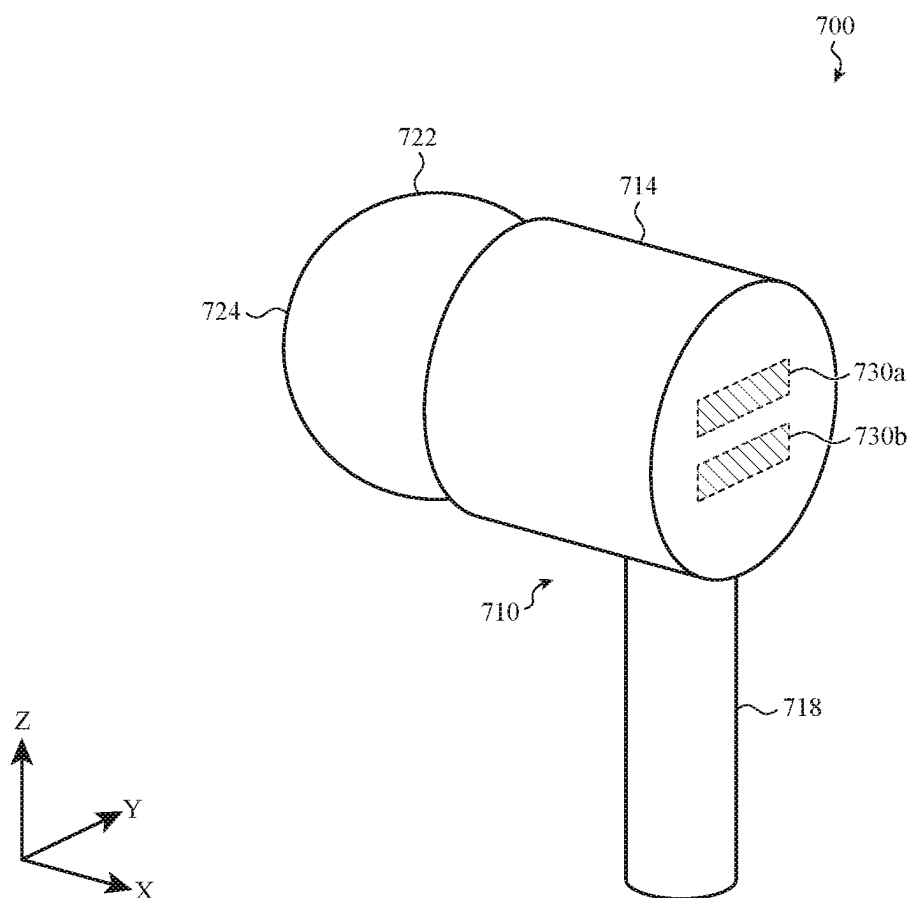


FIG. 7

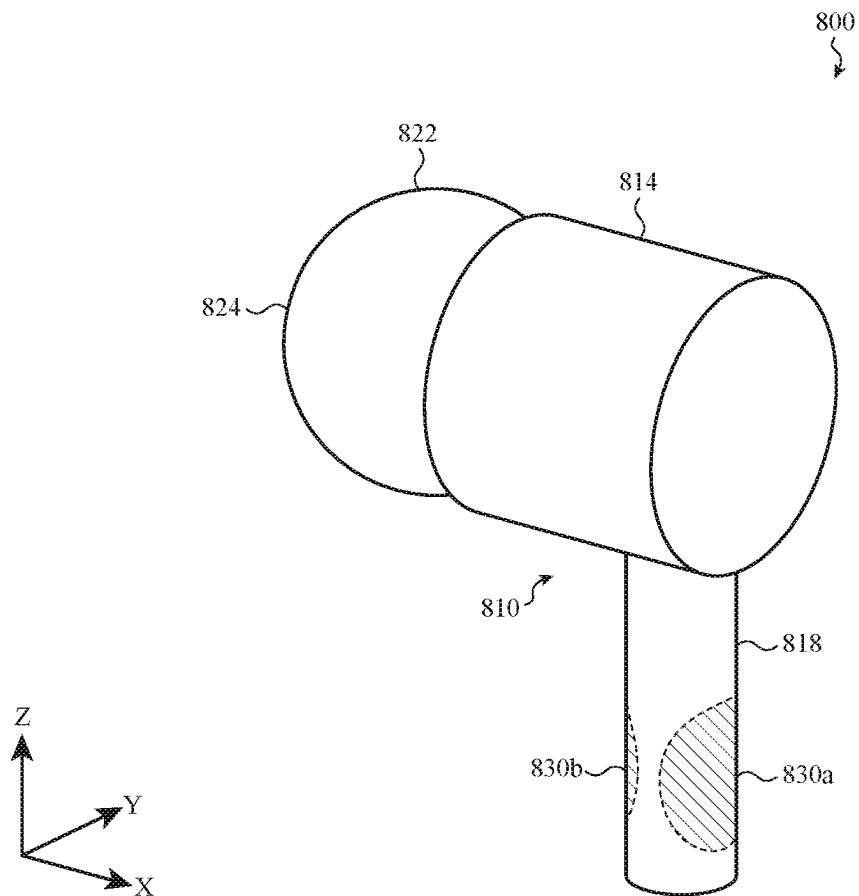


FIG. 8

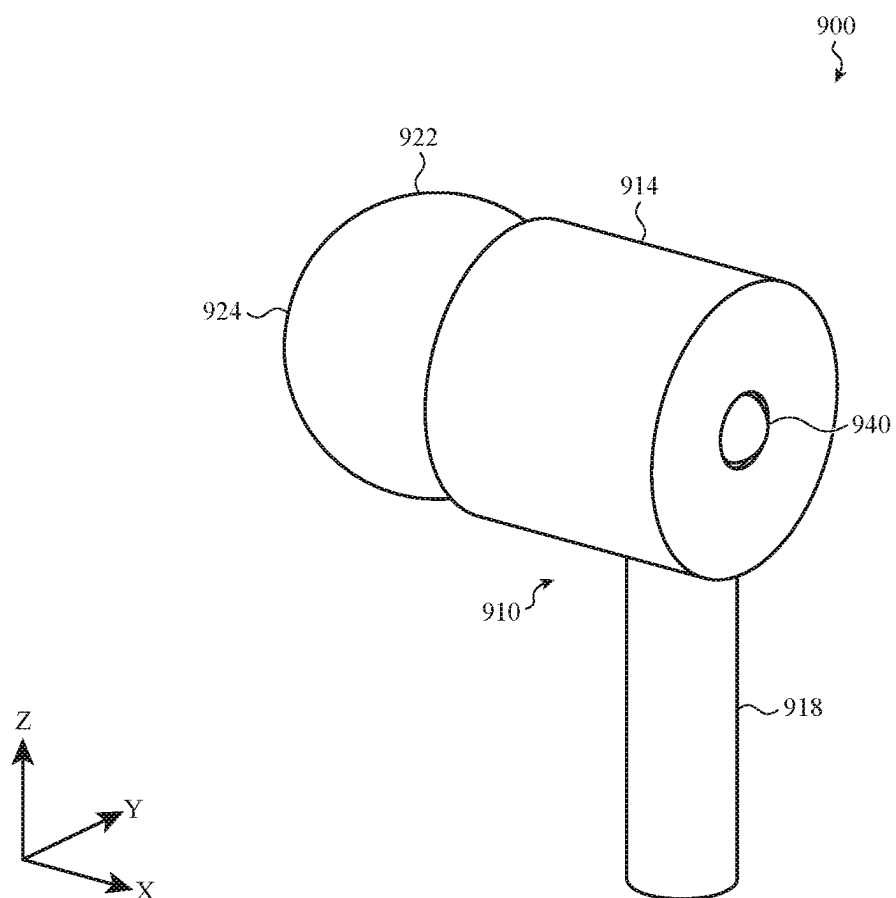


FIG. 9

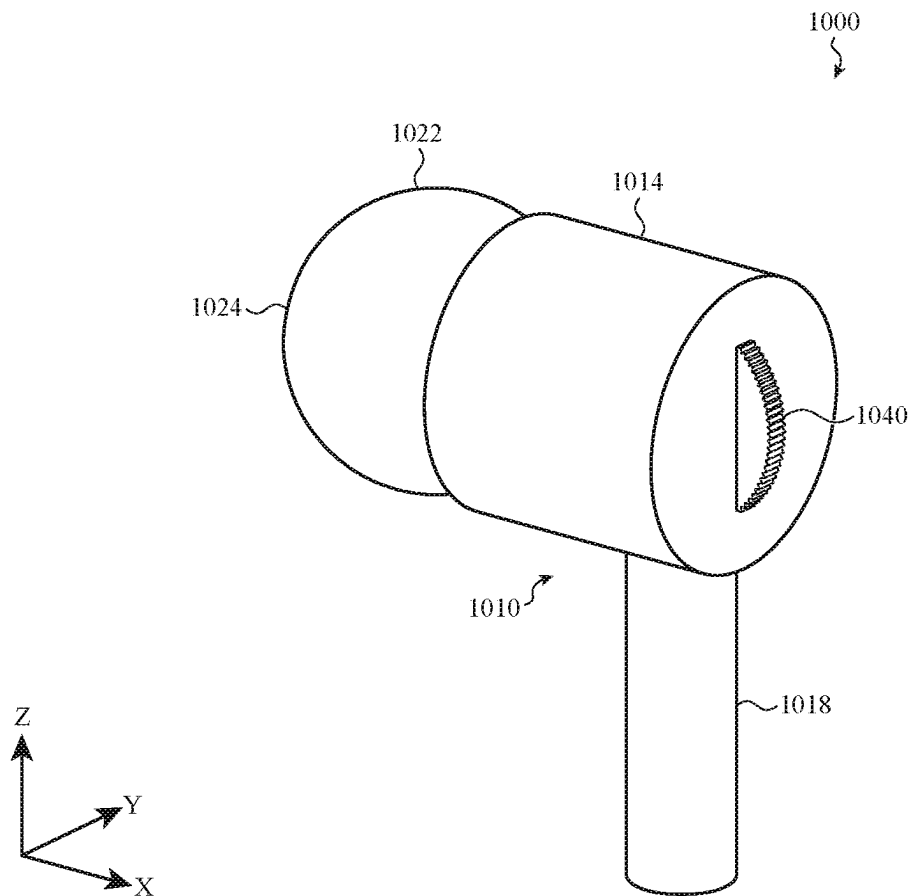


FIG. 10

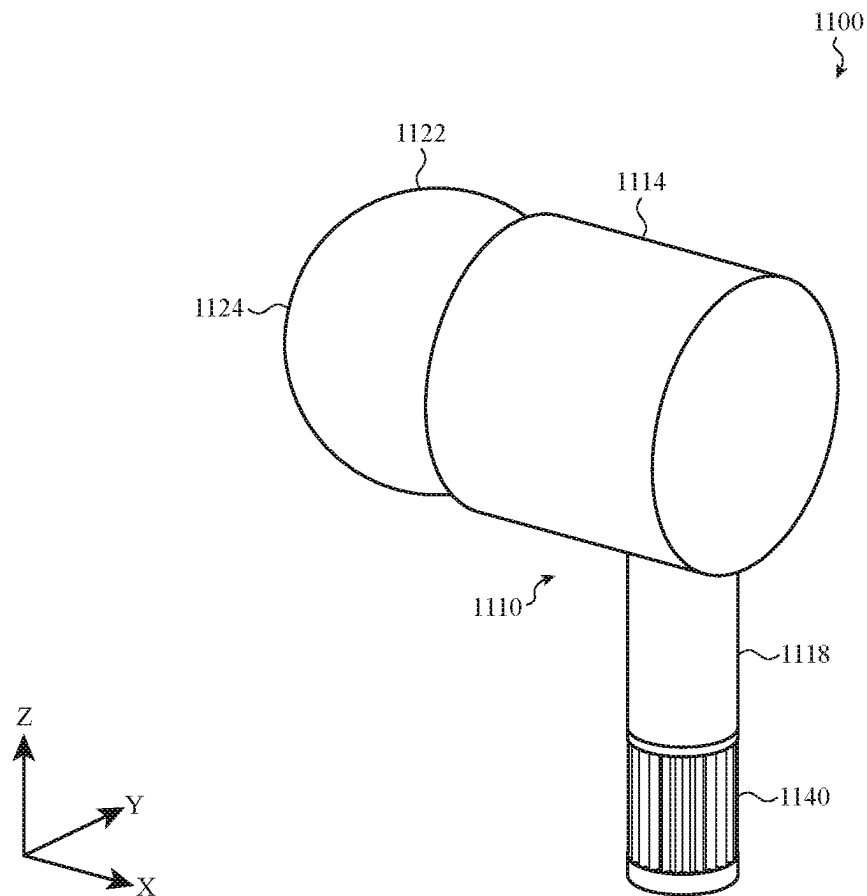


FIG. 11

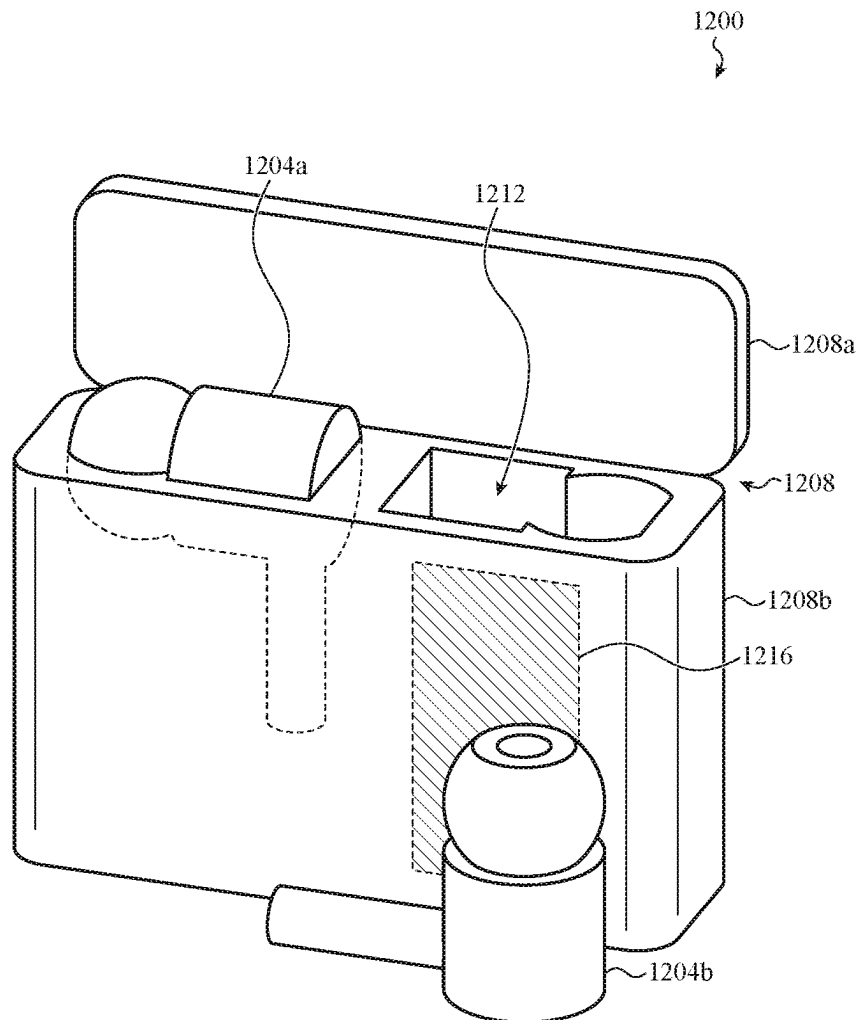
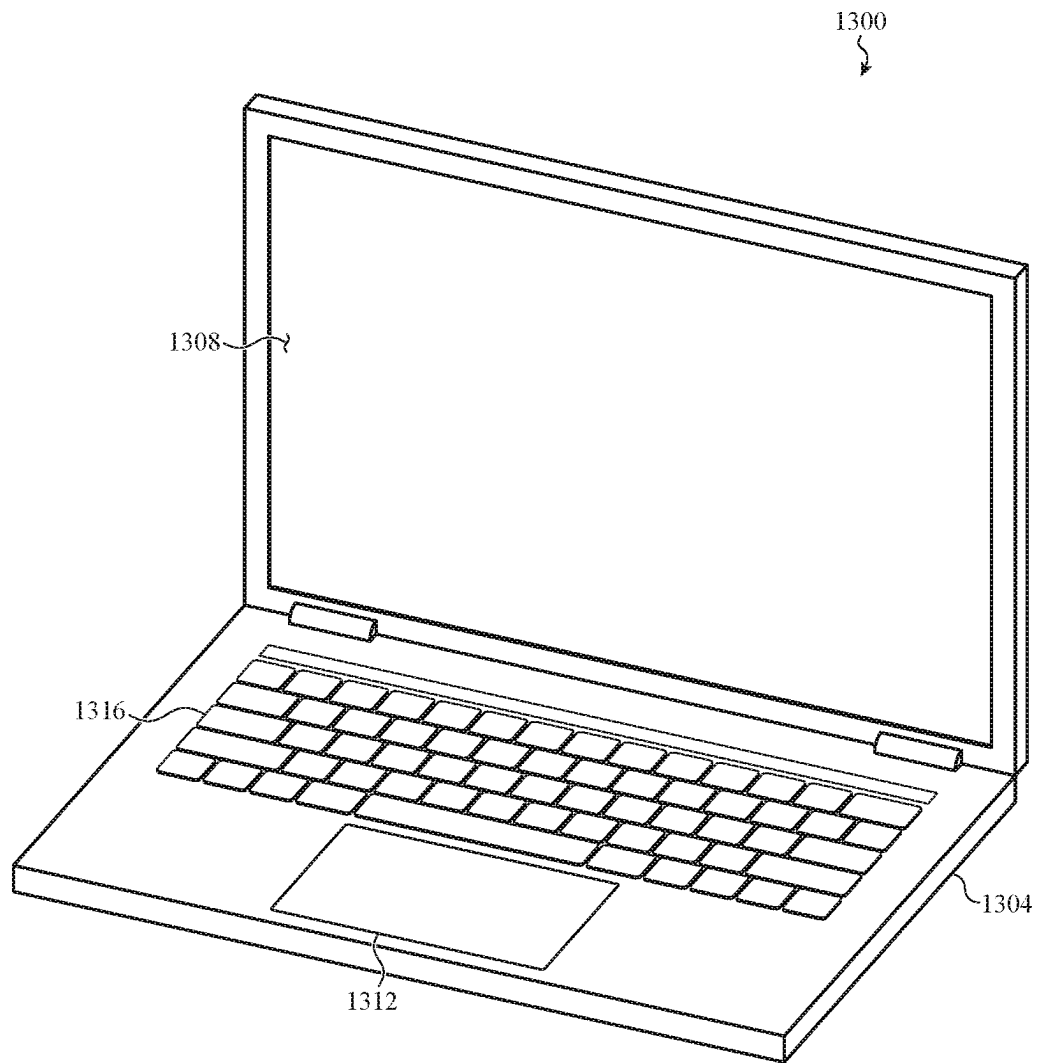
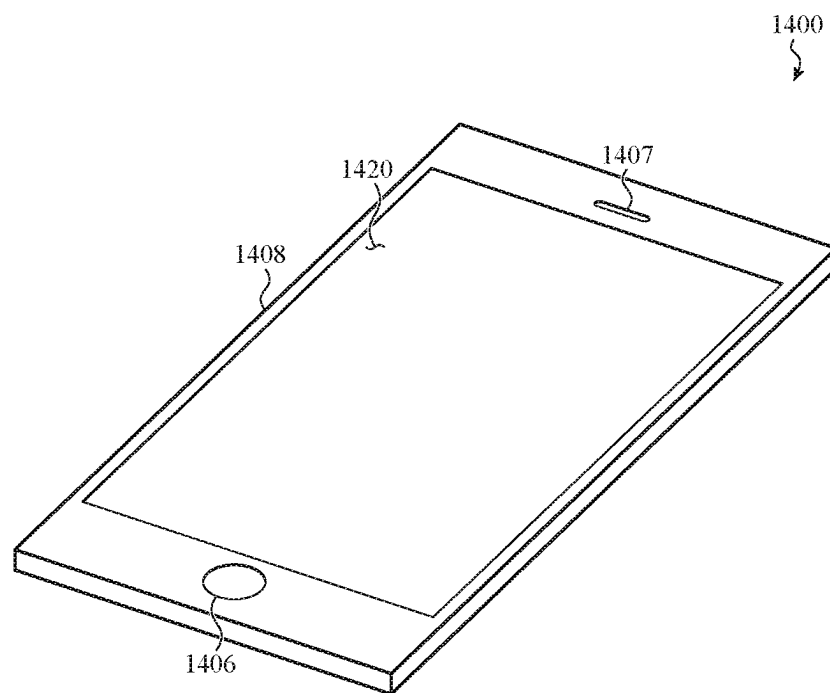
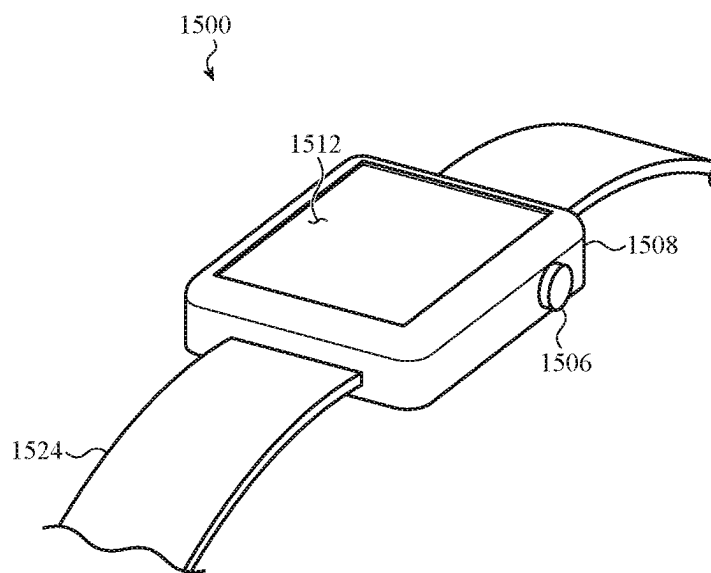


FIG. 12

*FIG. 13*

**FIG. 14**

*FIG. 15*

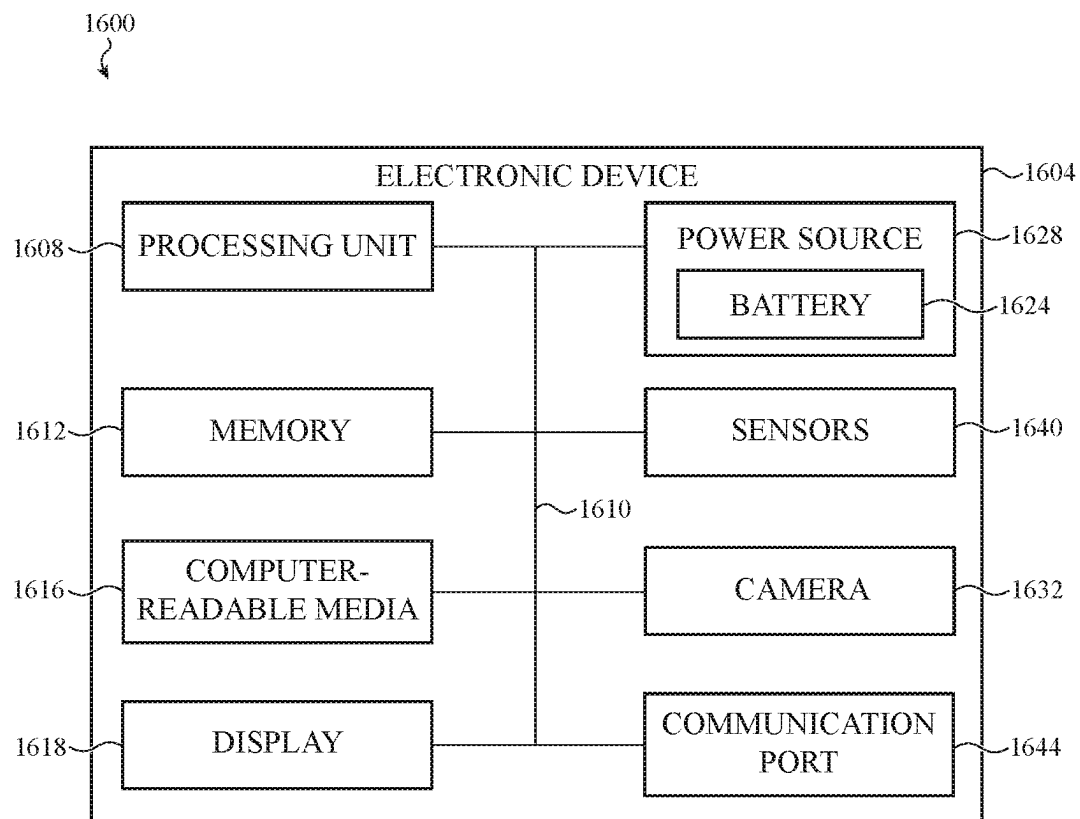


FIG. 16

WEARABLE INTERACTIVE AUDIO DEVICE**CROSS-REFERENCE TO RELATED APPLICATION**

This application is a continuation patent application of U.S. patent application Ser. No. 16/927,367, filed Jul. 13, 2020 and titled “Wearable Interactive Audio Device,” which is a continuation patent application of U.S. patent application Ser. No. 16/055,071, filed Aug. 4, 2018 and titled “Wearable Interactive Audio Device,” now U.S. Pat. No. 10,757,491, which is a nonprovisional patent application of and claims the benefit of U.S. Provisional Patent Application No. 62/683,594, filed Jun. 11, 2018 and titled “Wearable Interactive Audio Device,” the disclosures of which are hereby incorporated herein by reference in their entireties.

FIELD

Embodiments relate generally to wearable audio devices. More particularly, embodiments relate to sensors and structures that facilitate detection of input at a wearable audio device.

BACKGROUND

An earbud is worn at least partially inside of the ear of a user and typically is configured to produce a range of sounds based on a signal from another device. Many traditional earbuds suffer from significant drawbacks that may limit the ability to control sounds, or other outputs, at the earbud. In many cases, the earbud requires a hardwired connection that physically couples the earbud to another device and the sound is controlled based on input received at the device. Further, earbuds and/or other connected devices may be unresponsive to voice commands, thereby limiting the adaptability of the earbud to control multiple types of functions.

SUMMARY

Embodiments of the present invention are directed to a wearable interactive audio device.

In a first aspect, the present disclosure includes a wearable audio device. The wearable audio device includes an enclosure defining an opening. The wearable audio device further includes a sealing component connected to the enclosure and configured to define a sealed passage from the opening to an ear of a user. The wearable audio device further includes a speaker acoustically coupled to the sealed passage. The wearable audio device further includes a sensor positioned within the enclosure and configured to detect: (i) a direction of a first gesture input; and (ii) a direction of a second gesture input. The wearable audio device further includes a processing unit operatively coupled to the sensor and the speaker, the processing unit configured to, in response to the sensor detecting the direction of the first gesture input, cause the speaker to output a first audio signal. The processing unit may be further configured to, in response to the sensor detecting the direction of the second gesture input, cause the speaker to output a second audio signal.

In a second aspect, the present disclosure includes a wearable audio device. The wearable audio device includes an enclosure defining an earbud body. The wearable audio device further includes a sealing component connected to the enclosure and having a conformable surface. The wearable audio device further includes a tactile structure posi-

tioned on an exterior of the enclosure. The wearable audio device further includes a sensor positioned within the enclosure and configured to detect: (i) a first manipulation of the tactile structure; and (ii) a second manipulation of the tactile structure. The wearable audio device further includes a speaker configured to provide an audio output through the sealing component. The speaker may be configured to change the audio output in response to each of: (i) the first manipulation; and (ii) the second manipulation.

In a third aspect, the present disclosure includes a wearable audio device. The wearable audio device includes an enclosure defining an exterior surface of the wearable audio device and an opening. The wearable audio device further includes a sealing component positioned around the opening and configured to couple an interior volume of the enclosure with a user's ear canal. The wearable audio device further includes a first sensor positioned within the enclosure and configured to detect a gesture along the exterior surface. The wearable audio device further includes a second sensor positioned within the enclosure and configured to detect an audio input. The wearable audio device further includes a processing unit operatively coupled with the first sensor and the second sensor and configured to control a function of the wearable audio device using the audio input based on a detection of the gesture input.

In addition to the example aspects and embodiments described above, further aspects and embodiments will become apparent by reference to the drawings and by study of the following description.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like elements.

FIG. 1A depicts a functional block diagram of a wearable audio device;

FIG. 1B depicts a functional block diagram of a wearable audio device and a companion device;

FIG. 2 depicts sample wearable audio devices worn by a user;

FIG. 3A depicts a sample wearable audio device separated from a user;

FIG. 3B depicts a cross-sectional view of the sample wearable audio device and the user of FIG. 3A, taken along line A-A of FIG. 3A;

FIG. 4A depicts a sample wearable audio device connected to a user's ear;

FIG. 4B depicts a cross-sectional view of the sample wearable audio device of the user of FIG. 4A, taken along line B-B of FIG. 4A;

FIG. 5 depicts a sample wearable audio device worn by the user and the user's hand;

FIG. 6A depicts a sample wearable audio device having an input region;

FIG. 6B depicts a cross-sectional view of the sample wearable audio device of FIG. 6A, taken along line C-C of FIG. 6A;

FIG. 7 depicts a sample wearable audio device having multiple input regions;

FIG. 8 depicts a sample wearable audio device having an input region on a stem;

FIG. 9 depicts a sample wearable audio device having a button;

FIG. 10 depicts a sample wearable audio device having a rotary wheel;

FIG. 11 depicts a sample wearable audio device having a stem with a rotatable portion;

FIG. 12 depicts a sample wearable audio device and a case;

FIG. 13 depicts a sample notebook computer;

FIG. 14 depicts a sample portable device;

FIG. 15 depicts a sample watch; and

FIG. 16 depicts a functional block diagram of an electronic device.

The use of cross-hatching or shading in the accompanying figures is generally provided to clarify the boundaries between adjacent elements and also to facilitate legibility of the figures. Accordingly, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, element proportions, element dimensions, commonalities of similarly illustrated elements, or any other characteristic, attribute, or property for any element illustrated in the accompanying figures.

Additionally, it should be understood that the proportions and dimensions (either relative or absolute) of the various features and elements (and collections and groupings thereof) and the boundaries, separations, and positional relationships presented therebetween, are provided in the accompanying figures merely to facilitate an understanding of the various embodiments described herein and, accordingly, may not necessarily be presented or illustrated to scale, and are not intended to indicate any preference or requirement for an illustrated embodiment to the exclusion of embodiments described with reference thereto.

DETAILED DESCRIPTION

The description that follows includes sample systems, methods, and apparatuses that embody various elements of the present disclosure. However, it should be understood that the described disclosure may be practiced in a variety of forms in addition to those described herein.

The present disclosure describes systems, devices, and techniques related to a wearable audio device, such as an earbud or other device that may be worn at least partially in an ear of a user. The wearable audio device may form a sealed passage with an ear canal of the user when worn. A speaker of the wearable audio device may provide audio output through the sealed passage. The audio output may include music, voice communications, instructions, sounds, alerts, and so forth. The audio output may be initiated or controlled by a processing unit of the wearable audio device and/or an associated companion device, as described herein. The audio output may be responsive to various types of input received at the device, including touch and gesture inputs and physical manipulations of controls or other tactile structures.

The audio output may also be responsive to audio input, such as voice commands, received at the wearable audio device. The processing unit may distinguish between the various types of input in order to control one or more functions of the wearable audio device. For example, voice commands may be used to control music playback (e.g., play, pause, volume up/down), answer or end a call, interact with an intelligent digital assistant, and so forth. In some cases, voice commands may be used to control the device in response to detecting another type of input, such as a touch or a gesture. This may allow the wearable audio device to operate in various modes, including a first mode in which

audio input is temporarily ignored, and a second mode in which identified voice commands are used to control the device.

To facilitate the foregoing, the wearable audio device may include one or more sensors that detect different types of input. Broadly, the sensors may include substantially any input device, sensor, sensing element, sensing structure, switch, or the like, or combination thereof, that is responsive to environmental changes around the wearable audio device.

As one example, an incorporated sensor may detect a touch or proximity of a user or object to the wearable audio device. Continuing the example, a capacitive sensor may be incorporated into an enclosure of the wearable audio device and used to detect a capacitance between an electrode of the sensor and the user. As the user moves toward (and optionally presses on) the enclosure, the capacitance changes. This change (or absolute value of the capacitance) may be used to determine a position of a user's finger relative to the enclosure.

Likewise, an array of capacitive sensors may be operatively coupled to one another and used to track movement of the user along the enclosure. This may allow the wearable audio device to detect multiple gestures on or along the enclosure (e.g., a first gesture input, a second gesture input, and so on). For example, the sensor may detect a first direction of a gesture input and a second direction of a gesture input along the enclosure (or other directions, motions, and so on the gesture input). The processing unit of the wearable audio device may, in turn, initiate one or more functions of the wearable audio device based on the detected input, such as increasing a playback volume in response to the first direction of motion and/or decreasing playback volume in response to the second direction of motion, among other possible functions.

It will be appreciated that various types of sensors may be used to detect a touch, gesture, or the like at the enclosure and are described in greater detail below. For example, one or more optical sensors may be positioned within the enclosure and may detect a location of a user's hand, finger, or other object. An optical coating (transparent only to certain wavelengths) may substantially conceal the optical sensor from a user by camouflaging a lens and/or protective window of the sensor with the surrounding enclosure. Magnetic sensors, strain sensors, resistive sensors, and electroactive polymers (EAPs) may be used to detect such input at the wearable audio device, as described in greater detail below.

Additionally or alternatively, one or more tactile structures may be incorporated within or along an exterior of the enclosure and detect input at the wearable audio device. Broadly, a tactile structure may be substantially any physical feature of the device enclosure, including a button, switch, or other control, that may be physically manipulated by a user. Possible structures include textured surfaces of the enclosure, multi-input domes, rotary wheels and/or crowns, deformable stems, and so on. In some cases, the tactile structure may be configured to provide haptic feedback to the user in response to such manipulation, which may indicate receipt of input by the wearable audio device. Haptic feedback may be provided in part by forming the tactile structure from domes, beams, or other deformable materials and/or coupling the tactile structure with various electrically or thermally actuated haptic structures, as described herein.

Various sensors may be configured to detect manipulation of the tactile structure (e.g., a first manipulation, a second manipulation) and provide a corresponding signal that may be used to control, for example, the audio output and/or

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other global functions of the wearable audio device. For example, the sensors may detect a squeeze, a rotation, a swipe, a press, a jog, and so on of the tactile structure. To illustrate, where the tactile structure is a wheel, the sensors may detect a first rotation of the wheel and a second rotation of the wheel (or other physical manipulation), and the audio output may be responsive to each of the detected rotations. As another example, when the tactile structure is a textured surface of the enclosure, the sensor may detect a first swipe of the textured surface and a second swipe of the textured surface (or other manipulation), and the audio output may be responsive to each of the detected swipes. In this manner, the tactile structure may be used to indicate a boundary or region of the enclosure that may be touch-sensitive. For example, a user may tactilely perceive a change in surface texture between a surrounding device enclosure and the tactile structure, which signifies a region of the device that receives touch input. This may allow the user to accurately perform a gesture or other input on the enclosure while the wearable audio device is worn in the ear, or while the user is otherwise not visually perceiving the enclosure. In other cases, other tactile structures may be used and are described in greater detail herein, including embodiments in which multiple different tactile structures and/or combination of tactile structures and sensors are used to detect physical manipulations and/or gestures inputs at the enclosure.

For example, in an embodiment, the sensors may include one or more microphones positioned at least partially within the enclosure of the wearable audio device. As stated above, one microphone may detect voice commands that are used by the on-board processing unit to control, for example, the audio output of a speaker. This microphone may also be operatively coupled to another sensor (via the processing unit) so that such voice commands are accepted in conjunction or succession with another input, such as a touch input in a particular location on the device. Another microphone may detect other types of input, including input received by a body (e.g., head) of a user. For example, the microphone may detect an acoustic signal associated with a tap of a user's hand on a portion of the user's head, a click of a user's teeth, and/or other body movement that may be associated with input. This may allow the user to control audio output or another function of the device without necessarily physically contacting the wearable audio device and/or any associated companion device, thereby improving the adaptability of the system. This may also increase the interaction area that may receive input and reduce acoustic impacts of tapping the wearable audio device, among other improvements. Other microphones may be used and are described in greater detail below, including various beam forming and other microphones that may be used to detect and filter ambient noise.

The wearable audio device may include various structural components that house the sensors and tactile structures described herein. The enclosure of the wearable audio device may generally define an earbud body, which is shaped or otherwise contoured to be at least partially received within an ear of a user. The enclosure may have an interior volume that houses the sensors, speakers, microphones, processing unit(s), and other components and define an opening that allows the audio output from the speaker to exit. A sealing component, such as an elastic or deformable ring, may be positioned around the opening and configured to form a sealed passage between the interior volume and an ear canal of the user. To facilitate the forgoing, the sealing component may have a conformable surface that contacts

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the user's outer ear and engages, for example, the concha, the antihelix, the helix, or the like, in order to form the seal.

The enclosure may also have, or be formed from, various structural components that facilitate detection of input and/or receipt of the device within a user's ear. For example, in an embodiment, the enclosure may include a main unit and a stem. The main unit may define the opening through which audio output is provided to the user. The stem may extend from the main unit and include additional sensors, including touch sensors, and beam-forming microphones. The stem may generally be an elongated structure and, in some cases, house an antenna or other communications device. Both the main unit and the stem may be configured to receive a touch or gesture input and/or physical manipulation, as described herein. As a non-limiting illustration, the main unit may define a substantially uninterrupted external surface that may be used to receive a swipe of various directions or other gestures. As another illustration, the stem may be squeezed, rotated, or otherwise physically manipulated in order to provide input. It will be appreciated, however, that the main unit and the stem are presented herein for purposes of discussion, and that the enclosure of the wearable audio device may take substantially any shape to facilitate the various functions described herein.

Reference will now be made to the accompanying drawings, which assist in illustrating various features of the present disclosure. The following description is presented for purposes of illustration and description. Furthermore, the description is not intended to limit the inventive aspects to the forms disclosed herein. Consequently, variations and modifications commensurate with the following teachings, and skill and knowledge of the relevant art, are within the scope of the present inventive aspects.

FIG. 1A depicts a functional block diagram of a wearable audio device **100**, such as the wearable audio device discussed above and described in greater detail below. The wearable audio device **100** includes one or more input devices **110**, one or more output devices **140**, and a processing unit **150**. Broadly, the input devices **110** detect various types of input, and the output devices **140** provide various types of output. The processing unit **150** receives input signals from the input devices **110** in response to inputs detected at the input devices. The processing unit **150** may interpret input signals received from one or more input devices **110** and send output signals to one or more output devices **140** that instruct the output devices **140** to provide output. Detected input at one or more input devices **110** may be used to control one or more functions of the wearable audio device **100**. In this regard, the output devices **140** may be configured to provide outputs that may be manipulated based on the input detected at the input devices **110**. The outputs provided by the output devices **140** may also be responsive to, or initiated by, a program or application executed by a processing unit of the wearable audio device **100** and/or an associated companion device.

In various embodiments, the input devices **110** may include any suitable components for detecting inputs. Examples of input devices **110** include audio sensors (e.g., microphones), optical or visual sensors (e.g., cameras, visible light sensors, invisible light sensors), proximity sensors, touch sensors, force sensors, mechanical devices (e.g., switches, buttons, keys), vibration sensors, orientation sensors, motion sensors (e.g., accelerometers, velocity sensors), location sensors (e.g., GPS devices), thermal sensors, communication devices (e.g., wired or wireless communication devices), resistive sensors, magnetic sensors, electroactive polymers (EAPs), strain gauges, and so on, or some com-

bination thereof. Each input device **110** may be configured to detect one or more particular types of input and provide a signal (e.g., an input signal) corresponding to the detected input, for example to the processing unit **150**.

The output devices **140** may include any suitable components for providing outputs. Examples of output devices **140** include audio output devices (e.g., speakers), visual output devices (e.g., lights, displays), tactile output devices (e.g., haptic output devices), communication devices (e.g., wired or wireless communication devices), or some combination thereof. Each output device **140** may be configured to receive one or more signals (e.g., an output signal), for example from the processing unit **150**, and provide an output corresponding to the signal.

The processing unit **150** is operably coupled to the input devices **110** and the output devices **140**. The processing unit **150** is adapted to exchange signals with the input devices **110** and the output devices **140**. For example, the processing unit **150** may receive an input signal from an input device **110** that corresponds to an input detected by the input device. The processing unit **150** may interpret the received signal to determine whether to provide and/or change one or more outputs in response the input. The processing unit **150** may then send an output signal to one or more output devices **140** to provide and/or change outputs as appropriate. The processing unit **150** may include one or more computer processors or microcontrollers that are configured to perform operations in response to computer-readable instructions. Examples of suitable processing units are discussed in more detail below with respect to FIG. **16**.

As discussed above, in some embodiments, the input devices **110** include one or more microphones used to detect audio input. The audio input may include voice commands, vibrations, bodily noises, ambient noise, or other acoustic signals. In some cases, the wearable audio device **100** may have one or more dedicated microphones that are configured to detect particular types of audio input. For example, the wearable audio device **100** may include a first microphone, such as a beamforming microphone, that is configured to detect voice commands from a user, a second microphone that is configured to detect ambient noise, and a third microphone that is configured to detect acoustic signals or vibrations from a user's body (such as that produced by a facial tap or other gesture).

The processing unit **150** may receive a signal from each microphone and distinguish between the various types of input. For example, the processing unit **150** may identify a signal from the microphone(s) associated with an input (e.g., a voice command, a facial tap, and so on) and initiate a signal that is used to control a corresponding function of the wearable audio device **100**, such as an output provided by an output device **140**. The processing unit **150** may also identify signals from the microphone(s) associated with an ambient condition and ignore the signal and/or use the signal to control an audio output of the wearable audio device **100** (e.g., a speaker), such as acoustically cancelling or mitigating the effects of ambient noise.

One or more input devices **110** may operate to detect a location of an object or body part of a user relative to the wearable audio device **100**. This may also include detecting gestures, patterns of motion, signs, finger or hand positions, or the like, including detecting a direction of the input, such as a direction of a gesture along an x-axis, a y-axis, and/or a z-axis. To facilitate the foregoing, the wearable audio device **100** may include a capacitive sensor that is configured to detect a change in capacitance between an electrode of the sensor and a user. As the user approaches the sensor,

the capacitance changes, and thus may be used to determine a distance of the user relative to the electrode. In this manner, multiple capacitive sensors may be used to track a location or position of a body part of the user along an exterior surface of the wearable audio device **100**.

In some cases, the capacitive sensor may also be used to measure or detect a force input on an exterior surface of the wearable audio device **100**. For example, the user may press the exterior surface and deform it toward the electrode of the sensor. The surface may deform by a known amount for a given force, and thus a force applied by the user to the surface may be determined based on the positioned of the user derived from the change in capacitance.

As discussed above, the wearable audio device **100** may also include one or more visual or optical sensors. The optical sensors may, in certain embodiments, measure an intensity of light at one or more locations on the exterior surface of the wearable audio device **100**. A decrease in the intensity of light at a particular location may be associated with a user input or gestures, such as a cupping gesture over the wearable audio device **100**. A lens or protective window of the optical sensor may be camouflaged from a surrounding surface of the wearable audio device **100**, for example, using an optical coating, which may match the surrounding surface but be translucent to certain wavelengths of light. In other embodiments, the optical sensor may be, or form a component of, a camera or camera system. This may allow the wearable audio device **100** to detect and recognize specific types of gestures using pattern recognition.

Optical sensors, in certain embodiments, may also be used to detect a location of the wearable audio device **100**. For example, an optical sensor may be positioned relative to a portion of the wearable audio device **100** configured to be worn in a user's ear. This may allow the optical sensor to detect a receipt of the wearable audio device **100** within a person ear (e.g., in response to a decrease in light intensity measured at the sensor).

The input devices **110** may also include one or more mechanical devices or tactile structures that are configured to receive physical input or manipulations. Physical manipulations may include a squeeze, a collapse, a roll or rotation, a jog, a press, a pull, and so on. In some cases, the physical input may manipulate the mechanical device or tactile structure and cause the mechanical device or tactile structure to physically complete a switch or circuit that triggers a switch event. In other cases, the physical manipulation of the tactile structure is detected or recognized by substantially non-contact types of sensors or switches of the wearable audio device **100**, such as an optical reader detecting the rotation of a wheel, and so on. The mechanical device or tactile structure may therefore take various forms, including a textured exterior surface, a multi-input button or dome, a wheel, a crown, and so on.

The wearable audio device **100** may include various other components and sensors that are configured to detect input. In one embodiment, the wearable audio device **100** may include an antenna that is configured to communicatively or wirelessly couple the wearable audio device **100** to another device, such as the companion device **170** described below with respect to FIG. **1B**. Accordingly, the wearable audio device **100** may be configured to receive input signals from other devices such as the companion device **170**. As described above, the inputs may be used to control one or more outputs of the wearable audio device **100**, such as an audio output.

As a further example, the input devices **110** may include a thermal sensor to detect the placement of the wearable

audio device **100** within a user's ear. Accelerometers and speed sensors may be used to detect changing conditions, for example, when the wearable audio device **100** is used or otherwise worn by a user driving an automobile. In other cases, other combinations of sensors and associated functionalities are possible and contemplated herein.

As described above, an input device **110** may initiate or provide a signal corresponding to an input detected at the input device. The signal may be provided to the processing unit **150** and used to control one or more outputs of the wearable audio device **100**. In this regard, the wearable audio device **100** may include various output devices **140** in order to provide outputs and alter or manipulate the outputs based on detected inputs.

The output devices **140** may include one or more audio output devices, such as speakers, configured to produce an audio output, such as various types of music, voice communications, instructions, sounds, alerts, other acoustic signals, or some combination thereof. In some embodiments, the speakers have a relatively small form factor corresponding to that of the wearable audio device **100** so that the speakers may be disposed within an enclosure of the wearable audio device **100**. For example, the speaker may generally have a maximum dimension within a range of several millimeters, however other dimensions are possible. Notwithstanding, the speaker may be configured to provide substantially high resolution audio output to a user. This may be facilitated by the various components (e.g., sealing component **322** of FIG. 3A) described herein that are used define a sealed passage between an interior volume of the wearable audio device **100** (which houses the speaker) and the user's ear canal. The speaker may also be tuned to operate in one or more modes that facilitate canceling or mitigating ambient noise detected by, for example, one or more of the microphones of the wearable audio device **100**. For example, various characteristics of the audio output may be altered, for example by the processing unit **150**, in order to compensate for the interference of the ambient noise.

Audio outputs may be configured to change in response to inputs received at the wearable audio device **100**. For example, the processing unit **150** may be configured to change the audio output provided by a speaker in response to an input corresponding to a gesture input, physical manipulation, voice command, and so on. The speaker may thus receive multiple distinct signals from the processing unit **150** corresponding to different types of input or otherwise corresponding to distinct functions. To illustrate, a first signal corresponding to a direction of a first gesture input may cause the processing unit **150** to alter the audio output in a first manner (e.g., such as increasing playback volume in response to an up swipe), and a second signal corresponding to a direction of a second gesture input may cause the processing unit **150** to alter the audio output in a second manner (e.g., such as decreasing playback volume in response to a down swipe), among other possibilities.

The output devices **140** may include one or more tactile output devices configured to produce a tactile or haptic output. Haptic outputs may be facilitated by a haptic feedback structure, such as a dome, electromechanical actuator, and so forth. The output devices **140** may include one or more tactile structures to provide a tactile indication of, for example, the receipt of input by the wearable audio device **100**. This may include a buckling of a collapsible dome, or other deformation of a structure that registers input in response to a physical manipulation. Additionally or alternatively, a tactile structure may visually and/or tactilely indicate a region of the wearable audio device **100** operable

to receive input. For example, a textured surface may provide a tactile output to a user as the user feels the changing contours of the surface.

The output devices **140** may include one or more visual output devices configured to illuminate or otherwise visually alter a portion of the wearable audio device **100**, such as an exterior surface. Various lights or visual indicators may be used to produce a visual output of the wearable audio device **100**. The visual output may be indicative of an operational status of the wearable audio device **100**. For example, the visual output may include certain colors that represent a power-on mode, a standby mode, a companion-device pairing mode, a maintenance mode, and so on.

Visual output may also be used to indicate a receipt of input by the wearable audio device **100**. As one possibility, visual indicators along a surface of the wearable audio device **100** may produce a momentary flash, change colors, and so on, in response to received inputs. In this regard, the visual output may be responsive or adaptable to the various different types of input detected or that otherwise correspond to distinct functions of the wearable audio device **100**. This may include producing a first visual output (e.g., a first color, animation, or sequence) in response to a first input (audio, gesture, mechanical, and so forth) and producing a second visual output (e.g., second color, animation, or sequence) in response to a second input (audio, gesture, mechanical, and so forth).

Additional or alternative output devices **140** may generally be configured to produce other types of output, including, but not limited to, thermal outputs, pressure outputs, outputs for communication to external or companion devices, and so on. In one embodiment, the wearable audio device **100** may include an antenna that is configured to communicatively or wirelessly couple the wearable audio device **100** to another device, such as the companion device **170** described below with respect to FIG. 1B. The wearable audio device **100** may thus transmit an output signal from the to the companion device **170** that may be used to control one or more outputs of the companion device **170**, such as an audio output.

The input devices **110** and the output devices **140** described with respect to FIG. 1A may include a collection of mechanical components, sensors, instruments, processing unit(s), computer-readable instructions, and so forth that collectively operate to perform the functions described herein. Rather than define discrete or isolated systems, it will be appreciated that the devices may use common or overlapping components to perform the described functions. Further, in addition to those described with respect to FIG. 1A, the wearable audio device **100** may include any other appropriate hardware (e.g., sensors, switches, antennas, processing units, memories), software (e.g., applications, system programs, engines), network components (e.g., communication paths, interfaces, routers), and so forth for use in facilitating any operations disclosed herein, for example, such as those described below with respect to FIG. 16.

FIG. 1B depicts a functional block diagram of a wearable audio device (such as the device of FIG. 1A) and a companion device **170**. In particular, the wearable audio device **100**, described above with respect to FIG. 1A, is shown communicatively coupled with a companion device **170**. The companion device **170** may be substantially any computing device that is configured to receive input and initiate a signal that is used to control the wearable audio device **100**. In some embodiments, the functionality of the companion device **170** is provided by the wearable audio device **100**. Sample companion devices include, but are not limited

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to, a personal computer, a notebook computer, a tablet, a smart phone, a watch, a case for the wearable audio device **100**, a home automation device, and so on. Example companion devices, and corresponding structures are described herein, for example, with respect to FIGS. **12-16**.

The wearable audio device **100** and the companion device **170** may be communicatively coupled via a wireless connection. For example, the wearable audio device **100** may be paired with the companion device **170** using a short range wireless interconnection; however, other wireless connection techniques and protocols may be used. In other embodiments, the wearable audio device **100** and the companion device **170** may be connected via a wired connection.

FIG. **1B** depicts various functional modules of the companion device **170**. Each functional module or submodule described with respect to FIG. **1B** may include a collection of mechanical components, sensors, instruments, processing unit(s), computer-readable instructions, and so forth that collectively operate to perform the functions described herein. It will be appreciated that the companion device **170** also may include any appropriate hardware, software, network components, and so forth for use in facilitating any operations disclosed herein, for example, such as those described below with respect to FIG. **16**.

For purposes of illustration, the companion device **170** includes at least a context module **175**, an input module **180**, and an output module **185**. Broadly, the context module **175** may be configured to provide an operational context to the wearable audio device **100**. An operational context may be, or include, information associated with an application or program executed on the companion device **170** (e.g., such as an application executed by the processing unit **1608** of FIG. **16**). The operational context may therefore be used by the wearable audio device **100** to provide an output, such as a music output (where the executed program is an audio file), a voice communication output (where the executed program is a telephone call), an audio notification output (where the executed program is a navigation application), among various other possibilities.

The operational context may also be used by the wearable audio device **100** to determine or activate a particular type of input or sensor. For example, different types of gestures, audio input, physical manipulations, and so forth may be registered as input (or ignored) based on the operational context. To illustrate, where the operational context causes the wearable audio device **100** to output music, the processing unit **150** may be configured to control the music based on a direction of motion of a received input (e.g., such as a swipe in a first direction, a second direction, and so on). In another mode, where the operational context causes the wearable audio device **100** to output voice communications, the processing unit **150** may be configured to control the music based on a physical manipulation of a tactile structure (and ignore gesture inputs), among various other possibilities.

With reference to the input module **180**, the companion device **170** may be configured to receive input using various different sensors and structures. For example, and as described in greater detail below with respect to the sample companion devices of FIGS. **12-16**, the companion device **170** may include mechanical buttons, keyboards, touch-sensitive surfaces, trackpads, microphones, and other sensors. The input detected by the input module **180** may be used to control an output of the wearable audio device **100**. As one example, an audio playback volume may be increased or decreased in response to a manipulation of one or more mechanical keys or buttons of the companion device

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170. The input detected by the input module **180** may also be used to control a mode of the wearable audio device **100**, such as a mode for detecting certain audio inputs. For example, the wearable audio device **100** may be configured to enter a mode in which audio input is used to control a function of the wearable audio device **100** and/or the companion device **170**.

With reference to the output module **185**, the companion device **170** may be configured to provide output using various different components and structures. For example, and as described in greater detail below with respect to the sample companion devices of FIGS. **11-16**, the companion device **170** may include speakers, a display, tactile structures, and other components. The output provided by the output module **185** may be responsive to input detected by the wearable audio device **100**. As one example, in response to a detection of input at the wearable audio device **100**, a graphic may be depicted at a display of the companion device **170**, or, likewise, a sound may be produced at a speaker of the companion device **170**. The output module **185**, more generally, may also be used to indicate a status of the wearable audio device **100** to a user. For example, the output module **185** may produce an output, visual or otherwise, corresponding to different modes of the wearable audio device **100**, including a power-on mode, a standby mode, a battery status level, among various other indications.

FIG. **2** depicts wearable audio devices **200** worn by a user **250**. The wearable audio devices **200** may be a pair of devices, such as a first wearable audio device **200a** and a second wearable audio device **200b**, shown in FIG. **2**. Each of the first wearable audio device **200a** and the second wearable audio device **200b** may be substantially analogous to the wearable audio device **100** described above with respect to FIGS. **1A** and **1B**. For example, the wearable audio devices **200** may each be configured to detect input (e.g., audio, gesture, physical manipulation, and so on) and alter an audio or other output based on the detected input. In this regard, it will be appreciated that the wearable audio device **200** may include similar components and/or be configured to perform similar functions as the wearable audio device **100** described above. Redundant explanation of these components is omitted here for clarity.

The wearable audio devices **200** may be worn by the user **250**, as shown in FIG. **2**. In particular, the wearable audio devices **200** may be configured to be received and temporarily secured at an outer ear of the user **205**. In some cases, at least a portion of each of the wearable audio devices **200** may engage or seal a portion of a user's ear (such as forming a seal around an ear canal). This may support the device within the ear, and optionally form a sealed passage between various audio output components of the device and an ear canal of the user **250**. In the embodiment of FIG. **2**, the first wearable audio device **200a** may be positioned at least partially within a first ear **260a** of the user **250**, and the second wearable audio device **200b** may be positioned within a second ear **260b** of the user **250**. While FIG. **2** shows the user **250** wearing both the first wearable audio device **200a** and the second wearable audio device **200b**, it will be appreciated that in some cases a user **250** may optionally wear a single one of the first wearable audio device **200a** or the second wearable audio device **200b**. Further, notwithstanding the wearable audio devices **200** being configured to form a sealed passage with an ear canal of the user **250**, each of the wearable audio devices **200** may be selectively removable by the user **250**, and therefore

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allow the user **250** to wear and remove the wearable audio devices **200** as needed or desired.

FIG. 2 also shows the user **250** holding a companion device **290**. The companion device **290** may be substantially analogous to the companion device **170** described above with respect to FIG. 1B. The companion device **290** may be wirelessly coupled with the wearable audio devices **200**. In some cases, the companion device **290** may be configured to transmit information or commands associated with an operational context to one or both of the wearable audio devices **200**. This may include information corresponding to an application or program executed on the companion device **290**. For example, the companion device **290** may execute a music playback program, and thus the companion device **290** may provide information to the wearable audio devices **200** corresponding to the playback. The wearable audio device **200** may receive the information associated with the operational context and provide an output to the user **250** based on the received information. For example, upon receipt of the music playback information, the wearable audio devices **200** may be configured to provide an audio output to the user **250** that is, or includes, the music playback executed by the companion device **290**.

The wearable audio device **200** may also detect input. This may be used to control, for example, a function of the wearable audio device **200** (e.g., as indicated by a manipulation of an output of the device) or a function of the companion device **290**. In some cases, each of the wearable audio devices **200** may detect the same type of input and/or otherwise be used to redundantly control the same function. This may be the case, for example, where each of the wearable audio devices **200** is used to control playback volume of an audio output in response to a swipe along an exterior surface of either device. Additionally or alternatively, each of the wearable audio devices **200** may detect different types of input and/or be used to control different functions. This may be the case where, for example, the first wearable audio device **200a** is used to control a first function (e.g., play track) in response to an input, and the second audio device **200b** is used to control a second function (e.g., stop track) in response to another input. Such controls may be interchangeable, programmable, or otherwise based at least partially on the operational context provided by the companion device **290**. Further, while FIG. 2 shows the wearable audio device **200** and the companion device **290**, it will be appreciated that the wearable audio device **200** may operate independently from any companion device. For example, and as described herein with respect to FIG. 16, applications, programs, or the like may be executed exclusively on one or both of the first wearable audio device **200a** or the second wearable audio device **200b**, without necessarily pairing the devices to another external device. The wearable audio device **200** are shown as earbuds in FIG. 2 as one example. In various embodiments, the wearable audio device **200** may take different forms, including as all or part of a headband, lanyard, or other object. In some embodiments, the wearable audio device **200** may be a part of a headset, such as a virtual reality headset.

FIG. 3A depicts a wearable audio device **300** and a user **350**. The wearable audio device **300** is shown in FIG. 3A separated from the user **350**. In this regard, rather than being worn, the wearable audio device **300** is shown in FIG. 3 as occupying a position that allows the user **350** to readily advance the wearable audio device **300** toward an ear **360** of the user **350**. The ear **360** may generally include an ear surface **364**. The wearable audio device **300** may be configured to engage and form a seal with the ear surface **364**.

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This may allow the wearable audio device **300** to provide an audio output directly to the user **350**, such as through a sealed passage defined between the audio component of the wearable audio device **300** and the user **350**.

To facilitate the foregoing, FIG. 3A shows a sample construction of the wearable audio device **300**. In particular, FIG. 3A shows the wearable audio device **300** as including an enclosure **310** and a sealing component **322**. The enclosure **310** may define an interior volume containing various sensors, output devices, and other components of the wearable audio device **300**. The enclosure **310** may have a form factor that allows the wearable audio device **300** to fit at least partially in the ear **360** of the user **350**. While the enclosure **310** may take many forms, in certain embodiments, such as that shown in FIG. 3A, the enclosure **310** may include a main unit **314** and a stem **318**. Each of the main unit **314** and the stem **318** may house the input and/or output components of the wearable audio device **300**, as appropriate for a given application. In one embodiment, the main unit **314** may house relatively larger components of the wearable audio device **300**, such as a speaker or processing unit, and the stem **318** may house components of the wearable audio device **300** that may benefit from the elongated shape of the stem **318**, such as various microphones, antennas, and so on.

The enclosure **310** may be connected to the sealing component **322**. In some embodiments, the sealing component **322** may be fitted or positioned around a side of the enclosure **310**. For example, the enclosure **310** may define a speaker opening and the sealing component **322** may be positioned around this opening. The sealing component **322** may be connected to the enclosure **310** and configured to define a sealed passage from the opening to an ear of a user. As such, the sealing component **322** may be permanently affixed to the enclosure **310**. In other cases, as described herein, the sealing component **322** may be connected to the enclosure **310** in a manner that allows the sealing component to be removed, adjusted, replaced, and so on. The use of the term “connected to” embraces both these options.

The sealing component **322** may be configured to engage the ear **360** of the user **350**. For example, the sealing component **322** may include a conformable surface **324** that may be pressed into the ear surface **364**. In some cases, this may form or define a sealed passage between internal acoustic components of the wearable audio device **300** and, for example, an ear canal of the user **350**. Accordingly, the sealing component **322** may be formed from a variety of materials, including elastically deformable materials, such as silicon, rubber, nylon, and various other synthetic or composite materials. As described herein, the sealing component **322** may, in some embodiments, be removable from the enclosure **310** by the user **350**. This may allow the user **350** to interchange various different sealing components with the enclosure **310** of the wearable audio device **300** based on user customizable preferences.

FIG. 3B depicts a cross-sectional view of the ear **360** of the user **350**, taken along line A-A of FIG. 3A. Broadly, the ear **360** may include an outer ear region **370**, a middle ear region **372**, and an inner ear region **374**. FIG. 3B shows various features of the ear **360** that may be coupled with the wearable audio device **300** when the wearable audio device **300** is worn by the user **350**. For example, features of the outer ear region **370** may include a helix **376**, an antihelix **378**, a concha **380**, an auricular lobule **382**, and an ear canal **384**, among other features. The ear canal **384** may extend from an external opening of the outer ear region **370**, through the middle ear region **372**, and towards the inner ear

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region 374 where acoustic signals from the wearable audio device 300 may be processed by the user 350.

FIG. 4A depicts the wearable audio device 300 and the user 350. In particular, FIG. 4A depicts a configuration in which the wearable audio device 300 is worn by the user 350. For example, the wearable audio device 300 may be received at least partially within the ear 360 of the user 350. When the wearable audio device 300 is worn by the user 350, the sealing component 322 may be pressed into the ear 360, thereby causing the conformable surface 324 of the sealing component 322 to contact and conform or partially conform to the ear surface 364. This may allow the sealing component 322 to form a sealed passage between various audio output components of the wearable audio device 300 and, for example, an ear canal of the user.

FIG. 4B depicts a cross-sectional view of the ear 360 of the user 350, taken along line B-B of FIG. 4A, and the worn wearable audio device 300. As shown in FIG. 4B, the wearable audio device 300 may be at least partially received with the ear 360. For example, the sealing component 322 may be received within a portion of the outer ear region 370. The sealing component 322 may contact various features of the outer ear region 370 and form a seal about the ear canal 384. For example, the sealing component 322 may contact one or more of the helix 376, the antihelix 378, the concha 380, and/or various other features of the outer ear region 370 that may be positioned about the ear canal 384.

As described herein, the sealing component 322 may form or facilitate a sealed passage between various internal components of the wearable audio device 300 and the ear canal 384 of the user 350. FIG. 4B shows a sealed passage 330. The sealed passage 330 may extend from an interior volume 311 of the enclosure 310 to the ear canal 384. The sealed passage 330 may allow the wearable audio device 300 to propagate audio outputs into the ear canal 384, while mitigating or preventing the audio output from being released into a surrounding environment. For example, FIG. 4B shows a speaker 334 providing an acoustic signal A1 through the sealed passage 330 and into the ear canal 384. This may improve sound quality and also allow the sealing component 322 to block ambient noises or other environmental containments from entering the ear canal 384.

FIG. 5 depicts a wearable audio device 500 worn by a user 550. The wearable audio device 500 may be substantially analogous to the wearable audio devices described herein. The wearable audio device 500 may therefore include similar components and/or perform similar functions of other wearable audio devices described herein, redundant explanation of which is omitted here for clarity.

The wearable audio device 500 is shown positioned partially within an ear 560 of the user 550. For example, the wearable audio device 500 may be at least partially received by one or more components of the ear 560 (e.g., such as an ear canal) and include one or more other portions that are positioned outside of the ear 560.

As described in greater detail below with respect to FIGS. 6A-11, the portions of the wearable audio device 500 positioned outside of the ear 560 may facilitate on-device control of the wearable audio device 500. In this regard, FIG. 5 shows a hand 570 proximate to the wearable audio device 500. The hand 570 may form various gestures along the wearable audio device 500 (e.g., a first gesture, a second gesture) which may be detected by one or more input devices of the wearable audio device 500. The gestures may be movements of the hand 570; and as such, the one or more input devices may detect a direction of the gesture input along, for example, an x-axis, a y-axis, and/or a z-axis. The

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hand 570 may also manipulate one or more structures of the wearable audio device 500 (such as a first or second manipulation of a tactile structure) which may also be detected by one or more input devices of the wearable audio device 500. In either case, the wearable audio device 500 may detect the input from the hand 570, or other object or body portion of the user 550, and control an output of the wearable audio device 500, such as controlling an audio output of a speaker.

FIG. 6A depicts a sample wearable audio device 600. The wearable audio device 600 may be substantially analogous to the wearable audio devices described herein. The wearable audio device 600 may therefore include similar components and/or perform similar functions of other wearable audio devices described herein, including an enclosure 610, a main unit 614, a stem 618, a sealing component 622, and a conformable surface 624, redundant explanation of which is omitted here for clarity.

The wearable audio device 600 shown in FIG. 6A includes a touch-sensitive region 630 that may be defined along substantially any exterior surface of the enclosure 610. For purposes of illustration, the touch-sensitive region 630 is defined on the exterior surface of the main unit 614. In other embodiments, the touch-sensitive region 630 may be defined on other portions of the main unit 614 and/or the stem 618 (e.g., as described below with respect to FIG. 8).

The touch-sensitive region 630 may be configured to detect various types of input, including gesture inputs performed along (e.g., on or above) the enclosure 610. For example, the wearable audio device 600 may include one or more sensors (such as any of the input devices 110 of FIG. 1A) that are positioned within, or partially within, the enclosure and configured to detect a direction of a first gesture input, a direction of a second gesture input, and so on performed along the touch-sensitive region 630. A processing unit of the wearable audio device 600 may receive input corresponding to the detected gesture and control one or more functions of the wearable audio device 600 such as an audio output, as described herein.

FIG. 6B depicts a cross-sectional view of the enclosure 610, the touch-sensitive region 630, and the sensor 628 of the wearable audio device 600 of FIG. 6A, taken along line C-C of FIG. 6A. As shown in FIG. 6B, the touch-sensitive region 630 may form part of an exterior wall or surface of the main unit 614 of the enclosure 610. Positioned within an interior 611 of the main unit 614 is a sensor 628. The sensor 628 may be substantially any of the sensors described with respect to the input devices 110 of FIG. 1A. For example, the sensor 628 may be capacitive-based and configured to detect a change in capacitance between an electrode and a user and/or other object that approaches the main unit 614 at the touch-sensitive region 630. Other possible sensors include optical sensors, which may detect change in an intensity of light along the input region 630 and/or otherwise detect optical changes (including pattern recognition, such as where the sensor 628 is a camera) that may correspond to input received at the wearable audio device 600.

In some embodiments, the sensor 628 may be a substantially non-contact-based sensor. For example, the sensor 628 may be concealed or camouflaged within the main unit 614 and may be actuated based on proximity of a user or object, rather than physical contact with the sensor. In other cases, the sensor 628 may be, or form, a component of a contact-based switch or other mechanism that is actuated by the physical contact from a user. For example, the sensor 628 may be a collapsible dome (as shown in FIG. 9), or other structure that includes electrical contacts that close in response to input or deformation on the enclosure 610.

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In the embodiment of FIG. 6B, the enclosure 610 includes a tactile structure 634. The tactile structure 634 may be any series or group of tactile features, ridges, bumps, grooves, discontinuities, and so on that may be tactilely distinguishable from an adjoining smooth or uninterrupted surface of the enclosure 610. The tactile structure 634 may be a textured surface used to indicate the presence of input functionality along the enclosure 610 of the wearable audio device 600. In this regard, the touch-sensitive region 630 may be defined on the tactile structure 634; however, this is not required.

The sensor 628 may detect various physical manipulations of the tactile structure 634. For example, the sensor 628 may detect a first manipulation of the tactile structure 634, a second manipulation of the tactile structure 634, and so forth and provide signals to an associated processing unit corresponding to the different manipulations. For example, the first manipulation may be a finger moving on the tactile structure 634 in a first direction and the second manipulation may be a finger moving on the tactile structure 634 in a second, different direction. In other cases, the manipulations may be one or more of a squeeze, a roll, a press, a jog, and so on. In one sample embodiment, as described in greater detail below, the tactile structure 634 may be a portion of a stem or other feature of the wearable audio device 600, and the sensor 628 may detect a corresponding manipulation.

FIG. 7 depicts a wearable audio device 700. The wearable audio device 700 may be substantially analogous to other embodiments described herein. The wearable audio device 700 may therefore include similar components and/or perform similar functions of other wearable audio device described herein, including an enclosure 710, a main unit 714, a stem 718, a sealing component 722, and a conformable surface 724, redundant explanation of which is omitted here for clarity.

The wearable audio device 700 shown in FIG. 7 may include multiple touch-sensitive regions, such as a first touch-sensitive region 730a and a second touch-sensitive region 730b. Each of the first touch-sensitive region 730a and the second touch-sensitive region 730b may be substantially analogous to the touch-sensitive region 630 described above with respect to FIGS. 6A and 6B. For example, each of the first touch-sensitive region 730a and the second touch-sensitive region 730 may be associated with one or more (or a common) sensor of the wearable audio device 700 that is configured to detect various gestures or manipulations performed along the enclosure 710. Redundant explanation of these components is omitted here for clarity.

The multiple touch-sensitive regions of FIG. 7 may be used to detect various different types of input that may be used to control different functions of the wearable audio device 700. For example, the first touch-sensitive region 730a may be associated with a first function of the wearable audio device 700 (e.g., such as a volume up function) and the second touch-sensitive region 730b may be associated with a second function of the wearable audio device 700 (e.g., such as a volume down function). The wearable audio device 700 may detect input at a particular touch-sensitive region and perform the associated function, thereby allowing the wearable audio device 700 to have multiple, distinct buttons defined on the surface. The functions may be programmable or otherwise customizable by the user, such that multiple different functions may be executed by the wearable audio device 700. This may be at least partially based on an operational context (such as that provided by the context module 175 of FIG. 1B), such as associating the touch-sensitive region with various different audio controls

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when the wearable audio device 700 is playing back an audio track, associating the touch-sensitive regions with various telecommunication controls when the wearable audio device 700 is handling a cellular call, and so on.

While the first touch-sensitive region 730a and the second touch-sensitive region 730b are shown as being defined on a substantially smooth exterior surface of the main unit 714, it will be appreciated that the multiple touch-sensitive regions may be defined on substantially any exterior surface of the enclosure 710. Some embodiments include touch-sensitive regions on both the main unit 714 and the stem 718 and/or multiple touch-sensitive regions on the stem 718 (e.g., as shown in FIG. 8). Further, it will be appreciated that each of the first touch-sensitive region 730a and the second touch-sensitive region 730b may include distinct tactile features or structures, such as structures that each produce a distinct tactile output when contacted. As such, the distinct tactile outputs may indicate both the presence of input functionality along the enclosure 710 and the presence of distinct input functionalities as between the respective touch-sensitive regions.

FIG. 8 depicts a wearable audio device 800 that is substantially analogous to the wearable audio devices described herein, in some respects. The wearable audio device 800 may therefore include similar components and/or perform similar functions of other wearable audio devices described herein, including an enclosure 810, a main unit 814, a stem 818, a sealing component 822, and a conformable surface 824, redundant explanation of which is omitted here for clarity.

The wearable audio device 800 shown in FIG. 8 may include multiple touch-sensitive regions, such as a first touch-sensitive region 830a and a second touch-sensitive region 830b. Each of the first touch-sensitive region 830a and the second touch-sensitive region 830b may be substantially analogous to the touch-sensitive region 630 described above with respect to FIGS. 6A and 6B. For example, each of the first touch-sensitive region 830a and the second touch-sensitive region 830b may be associated with one or more (or a common) sensors of the wearable audio device 800 that are configured to detect various gestures or manipulations performed on the enclosure 810. Redundant explanation of these components is omitted here for clarity.

The multiple touch-sensitive regions of the wearable audio device 800 may be defined on a portion of the stem 818 of the enclosure 810. For example, as shown in FIG. 8, the first touch-sensitive region 830a may be defined on an end of the stem 818 and the second touch-sensitive region 830b may be defined on the same end of the stem 818 opposite the first touch-sensitive region 830a.

The arrangement of the first touch-sensitive region 830a and the second touch-sensitive region 830b on the stem 818 may allow the stem 818 to receive various types of input. For example, the first touch-sensitive region 830a and the second touch-sensitive region 830b may cooperate to detect a squeeze, pinch, pull, press, and/or other manipulation of the stem 818. Continuing the example, the first touch-sensitive region 830a and the second touch-sensitive region 830b may each detect a touch that corresponds to a user contacting the stem 818 at the opposing sides. Upon a detection of the touch at each of the regions, a processing unit may control one or more functions of the wearable audio device 600, such as controlling an audio playback function, activate a voice command mode, and so on.

In some cases, the stem 818 may be or otherwise define a tactile structure. For example, the stem 818 may be a deformable structure that elastically bends, twists, and so on

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when contacted by a user. As such, the first touch-sensitive region **830a** and the second touch-sensitive region **830b** may cooperate to detect physical manipulations of the stem **818**. This may include, in certain embodiments, a first manipulation of the stem (e.g., such as a twist of the stem **818** in a first direction), a second manipulation of the stem (e.g., such as a twist of the stem **818** in a second, opposing direction), among various other types of physical manipulations.

FIG. **9** depicts a wearable audio device **900** that may be substantially analogous in some respects to the wearable audio devices described herein. The wearable audio device **900** may therefore include similar components and/or perform similar functions of other wearable audio devices described herein, including an enclosure **910**, a main unit **914**, a stem **918**, a sealing component **922**, and a conformable surface **924**, redundant explanation of which is omitted here for clarity.

The wearable audio device **900** shown in FIG. **9** may include one or more tactile structures positioned along an exterior of the enclosure **910**, such as tactile structure **940**. Tactile structure **940** may be a component of a mechanical switch or button that is configured to detect various types of input. For example, the tactile structure **940** may be a collapsible dome that buckles when pressed by a user. The collapsible dome may be a multi-input device, such that different signals are provided as a user presses into the dome (e.g., a first signal for a first manipulation or press of the dome, and a second signal for a second manipulation or press of the dome, and so on). In this regard, the tactile structure **940** may provide haptic feedback in response to the first manipulation and/or the second manipulation and close a corresponding switch or otherwise provide a signal corresponding to the respective manipulation. A processing unit may receive the provided signals and control a corresponding function of the wearable audio device **900**, such as controlling an audio output of the wearable audio device **900**.

FIG. **10** depicts a wearable audio device **1000**. The wearable audio device **1000** may be substantially analogous to the wearable audio devices described herein. The wearable audio device **1000** may therefore include similar components and/or perform similar functions of other wearable audio devices described herein, including an enclosure **1010**, a main unit **1014**, a stem **1018**, a sealing component **1022**, and a conformable surface **1024**, redundant explanation of which is omitted here for clarity.

The wearable audio device **1000** may include one or more tactile structures positioned along an exterior of the enclosure **1010**, such as the tactile structure **1040**. The tactile structure **1040** may be a component of a mechanical switch or button that is configured to detect various types of input. For example, the tactile structure **1040** may be a wheel that rotates when manipulated by a user. As shown in FIG. **10**, the wheel may be positioned on or extend through a surface of the main unit **1014**; however, in other embodiments, the wheel may be positioned on other surfaces, including on a surface of the stem **1018**. The wheel may be a multi-input device, such that various different signals are provided as a user causes the wheel to rotate (e.g., a first signal for a first rotation of the wheel and a second signal for a second rotation of the wheel, and so on). A processing unit may receive the provided signals and control a corresponding function of the wearable audio device **1000**, such as controlling an audio output of the wearable audio device **1000**.

Rotation may be detected using various techniques, such as using one or more of the input devices **110** described above with respect to FIG. **1A**. For example, one or more

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optical sensors positioned within the wearable audio device **1000** may detect rotation of the wheel using reflected light. In other cases, mechanical devices may be used, which may detect the rotation of the wheel using electrical, magnetic, or physical contacts disposed within the wearable audio device **1000**. The tactile structure **1040** may also provide haptic feedback in response to the rotation. This may be a resistive force that opposes rotational input. Additionally or alternatively, the tactile structure **1040** may produce an audible click.

FIG. **11** depicts a wearable audio device **1100**. The wearable audio device **1100** may be substantially analogous to the wearable audio devices described herein. The wearable audio device **1100** may therefore include similar components and/or perform similar functions of other wearable audio device described herein, including an enclosure **1110**, a main unit **1114**, a stem **1118**, a sealing component **1122**, and a conformable surface **1124**, redundant explanation of which is omitted here for clarity.

The wearable audio device **1100** shown in FIG. **11** may include one or more tactile structures positioned along an exterior of the enclosure **1110**, such as the tactile structure **1140**. The tactile structure **1140** may be a component of a mechanical switch or button that is configured to detect various types of input. For example, the tactile structure **1140** may be a rotatable portion of the stem **1118** that is configured to rotate when manipulated by a user. As shown in FIG. **11**, the rotatable portion may be positioned on an end of the stem **1118**; however, in other embodiments, the rotatable portion may be positioned on other surfaces or features, including features of the main unit **1114**. The rotatable portion may be a multi-input device, such that various different signals are provided as a user causes the rotatable portion to rotate (e.g., a first signal for a first rotation of the rotatable portion and a second signal for a second rotation of the rotatable portion, and so on). A processing unit may receive the provided signals and control a corresponding function of the wearable audio device **1100**, such as controlling an audio output of the wearable audio device **1100**.

The rotation may be detected using various techniques, using one or more of the input devices **110** described above with respect to FIG. **1A**. For example, one or more optical sensors positioned within the wearable audio device **1100** may detect rotation of the rotatable portion using reflected light. In other cases, mechanical rotary detection devices may be used, which may detect the rotation of the rotatable portion using electrical contacts disposed within the wearable audio device **1100**. The tactile structure **1140** may also be configured to provide haptic feedback in response to the rotation. This may be a resistive force that opposes rotational input. Additionally or alternatively, the tactile structure **1040** may produce an audible click.

FIG. **12** depicts an example case **1200**. The case **1200** may generally be configured to receive one or more wearable audio devices, such as the first wearable audio device **1204a** and the second wearable audio device **1204b** shown in FIG. **12**. In this regard, the case **1200** may be used to store and protect the first wearable audio device **1204a** and the second wearable audio device **1204b** when not in use.

For purposes of illustration, the case **1200** is shown as having an enclosure **1208**. The enclosure **1208** includes an upper portion **1208a** and a lower portion **1208b**. The upper portion **1208a** may pivot relative to the lower portion **1208b**, which may open the case **1200**. Defined within a least the lower portion **1208b** may be one or more receiving features **1212**. The receiving feature **1212** may be configured to

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receive respective ones of the first wearable audio device **1204a** and the second wearable audio device **1204b**. For example, in some embodiments, the receiving features **1212** may be contoured to a matching shape of the audio devices, which may allow the devices to be supported or snugly received within the enclosure **1208**. The receiving features **1212** may also include various electrical connectors, terminals, adaptors, and so on that may electrically recharge wearable audio devices that are received by the receiving features **1212**.

The case **1200** may also function as a companion device to the wearable audio devices described herein, such as the first wearable audio device **1204a** and/or the second wearable audio device **1204b** shown in FIG. **12**. In this regard, the case **1200** may include a processing unit and may therefore perform similar functions to the companion devices described herein, such as the companion device **170** of FIG. **1B**. Redundant explanation of these functions is omitted here for clarity.

In the embodiment of FIG. **12**, the case **1200** may be configured to detect input, such as a gesture performed along an exterior of the enclosure **1208**. For example, as shown in FIG. **12**, the lower portion **1208b** may include a touch-sensitive region **1216**. One or more sensors, such as substantially any of the sensors of the input device **110** of FIG. **1A**, may be positioned within the enclosure **1208** and configured to detect the input at the touch-sensitive region **1216**. In some cases, the touch-sensitive region **1216** may be defined at a tactile structure or textured surface of the enclosure **1208** and the sensors within the case may be configured to detect various manipulations of the tactile structure. The sensor may detect the gesture, manipulation, or other input and provide a corresponding signal to, for example, a processing unit of the case **1200** and/or the wearable audio devices **1204a**, **1204b**. The processing unit may receive the provided signals and control a corresponding function of the wearable audio devices **1204a**, **1204b**, such as controlling an audio output of the wearable audio devices **1204a**, **1204b**, among other functions.

FIG. **13** depicts an example electronic device **1300**. The electronic device **1300** may be or function as a companion device to the wearable audio devices described herein. The electronic device **1300** may therefore perform similar functions to the companion devices described herein, such as the companion device **170** of FIG. **1B**. Redundant explanation of these functions is omitted here for clarity.

For purposes of illustration, the electronic device **1300** is a notebook computer. In this regard, the electronic device **1300** is shown in FIG. **13** as having an enclosure **1304**, a display **1308**, one or more input/output members **1312**, and a keyboard assembly **1316**. It should be noted that the electronic device **1300** may also include various other components, such as one or more ports (e.g., charging ports, data transfer ports, or the like), additional input/output buttons, and so on. As such, the discussion of any electronic device, such as electronic device **1300**, or other companion devices, is meant as illustrative only.

FIG. **14** depicts an example electronic device **1400**. The electronic device **1400** may be or function as a companion device to the wearable audio devices described herein. The electronic device **1400** may therefore perform similar functions to the companion devices described herein, such as the companion device **170** of FIG. **1B**. Redundant explanation of these functions is omitted here for clarity.

For purposes of illustration, the electronic device **1400** is a mobile phone. In this regard, the electronic device **1400** is shown in FIG. **14** as having an enclosure **1408**, a touch-

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sensitive display **1420**, one or more input/output members **1406**, and a speaker **1407**. It should be noted that the electronic device may also include various other components, such as one or more ports (e.g., charging ports, data transfer ports, or the like), additional input/output buttons, and so on. As such, the discussion of any electronic device, such as electronic device **1400** is meant as illustrative only.

FIG. **15** depicts an example electronic device **1500**. The electronic device **1500** may be or function as a companion device to the wearable audio devices described herein. The electronic device **1500** may therefore perform similar functions to the companion devices described herein, such as the companion device **170** of FIG. **1B**. Redundant explanation of these functions is omitted here for clarity.

For purposes of illustration, the electronic device **1500** is a watch or other wearable electronic device. In this regard, the electronic device **1500** is shown in FIG. **15** as having an enclosure **1508**, a crown **1506**, a touch-sensitive display **1512**, and a band **1524**. The touch-sensitive display **1512** may be positioned in a first opening defined by the enclosure **1508** and the crown **1506** may be at least partially positioned in a second opening defined by the enclosure **1508**. The touch-sensitive display **1512** may be responsive to translational and rotational movement of the crown **1506**. For example, a visual output of the touch-sensitive display **1512** may be modified in a first manner in response to rotational movement of the crown **1506** and in a second manner in response to translational movement of the crown **1506**. It should be noted that the electronic device **1500** may also include various other components, such as one or more ports (e.g., charging ports, data transfer ports, or the like), additional input/output buttons, and so on. As such, the discussion of any electronic device, such as electronic device **1500**, is meant as illustrative only.

FIG. **16** is a functional block diagram **1600** of a sample electronic device **1604**. It will be appreciated, however, that the functional block diagram described herein of the electronic device **1604** may include components representative of substantially any other electronic devices, wearable audio devices, companion devices, or the like described herein. In this regard, the schematic representation in FIG. **16** may correspond to the electronic device **1604**. However, the schematic representation in FIG. **16** may also represent the other electronic devices, audio devices, companion devices, or the like described herein, for example, such as the wearable audio device **100** of FIG. **1A** and companion device **170** of FIG. **1B**. The electronic device **1604** may include any appropriate hardware (e.g., computing devices, data centers, switches), software (e.g., applications, system programs, engines), network components (e.g., communication paths, interfaces, routers), and the like (not necessarily shown in the interest of clarity) for use in facilitating any appropriate operations disclosed herein.

As shown in FIG. **16**, the electronic device **1604** may include a processing unit or element **1608** operatively connected to computer memory **1612** and computer-readable media **1616**. The processing unit **1608** may be operatively connected to the memory **1612** and computer-readable media **1616** components via an electronic bus or bridge (e.g., such as system bus **1610**). The processing unit **1608** may include one or more computer processors or microcontrollers that are configured to perform operations in response to computer-readable instructions. The processing unit **1608** may be a central processing unit of the stylus. Additionally or alternatively, the processing unit **1608** may be other processors within the device including application specific integrated chips (ASIC) and other microcontroller devices.

The memory **1612** may include a variety of types of non-transitory computer-readable storage media, including, for example, read access memory (RAM), read-only memory (ROM), erasable programmable memory (e.g., EPROM and EEPROM), or flash memory. The memory **1612** is configured to store computer-readable instructions, sensor values, and other persistent software elements. Computer-readable media **1616** may also include a variety of types of non-transitory computer-readable storage media, including, for example, a hard-drive storage device, a solid-state storage device, a portable magnetic storage device, or other similar device. The computer-readable media **1616** may also be configured to store computer-readable instructions, sensor values, and other persistent software elements.

In this example, the processing unit **1608** is operable to read computer-readable instructions stored on the memory **1612** and/or computer-readable media **1616**. The computer-readable instructions may adapt the processing unit **1608** to perform the operations or functions described above with respect to FIGS. 1A-15. The computer-readable instructions may be provided as a computer-program product, software application, or the like. It should be appreciated that, where the electronic device is a stylus, the processing unit **1608** may be located in an electronic device associated with the stylus, rather than the stylus itself. In such embodiments, data may be transmitted from the stylus to and from the electronic device, such that the processing unit in the electronic device may operatively control the stylus.

As shown in FIG. 16, the electronic device **1604** may also include a display **1618**. The display **1618** may include a liquid-crystal display (LCD), organic light emitting diode (OLED) display, light emitting diode (LED) display, or the like. If the display **1618** is an LCD, the display may also include a backlight component that can be controlled to provide variable levels of display brightness. If the display **1618** is an OLED or LED type display, the brightness of the display **1618** may be controlled by modifying the electrical signals that are provided to display elements.

The electronic device **1604** may also include a battery **1624** that is configured to provide electrical power to the components of the electronic device **1604**. The battery **1624** may include one or more power storage cells that are linked together to provide an internal supply of electrical power. In this regard, the battery **1624** may be a component of a power source **1628** (e.g., including a charging system or other circuitry that supplies electrical power to components of the electronic device **1604**). The battery **1624** may be operatively coupled to power management circuitry that is configured to provide appropriate voltage and power levels for individual components or groups of components within the electronic device **1604**. The battery **1624**, via power management circuitry, may be configured to receive power from an external source, such as an AC power outlet or interconnected computing device. The battery **1624** may store received power so that the electronic device **1604** may operate without connection to an external power source for an extended period of time, which may range from several hours to several days.

The electronic device **1604** may also include one or more sensors **1640** that may be used to detect a touch and/or force input, environmental condition, orientation, position, or some other aspect of the electronic device **1604**. For example, sensors **1640** that may be included in the electronic device **1604** may include, without limitation, one or more accelerometers, gyroscopes, inclinometers, or magnetometers. The sensors **1640** may also include one or more

proximity sensors, such as a magnetic hall-effect sensor, inductive sensor, capacitive sensor, continuity sensor, or the like.

The sensors **1640** may also be broadly defined to include wireless positioning devices including, without limitation, global positioning system (GPS) circuitry, Wi-Fi circuitry, cellular communication circuitry, and the like. The electronic device **1604** may also include one or more optical sensors, including, without limitation, photodetectors, photosensors, image sensors, infrared sensors, or the like. In one example, the sensor **1640** may be an image sensor that detects a degree to which an ambient image matches a stored image. As such, the sensor **1640** may be used to identify a user of the electronic device **1604**. The sensors **1640** may also include one or more acoustic elements, such as a microphone used alone or in combination with a speaker element. The sensors **1640** may also include a temperature sensor, barometer, pressure sensor, altimeter, moisture sensor or other similar environmental sensor. The sensors **1640** may also include a light sensor that detects an ambient light condition of the electronic device **1604**.

The sensor **1640**, either alone or in combination, may generally be a motion sensor that is configured to estimate an orientation, position, and/or movement of the electronic device **1604**. For example, the sensor **1640** may include one or more motion sensors, including, for example, one or more accelerometers, gyroscopes, magnetometers, optical sensors, or the like to detect motion. The sensors **1640** may also be configured to estimate one or more environmental conditions, such as temperature, air pressure, humidity, and so on. The sensors **1640**, either alone or in combination with other input, may be configured to estimate a property of a supporting surface, including, without limitation, a material property, surface property, friction property, or the like.

The electronic device **1604** may also include a camera **1632** that is configured to capture a digital image or other optical data. The camera **1632** may include a charge-coupled device, complementary metal oxide (CMOS) device, or other device configured to convert light into electrical signals. The camera **1632** may also include one or more light sources, such as a strobe, flash, or other light-emitting device. As discussed above, the camera **1632** may be generally categorized as a sensor for detecting optical conditions and/or objects in the proximity of the electronic device **1604**. However, the camera **1632** may also be used to create photorealistic images that may be stored in an electronic format, such as JPG, GIF, TIFF, PNG, raw image file, or other similar file types.

The electronic device **1604** may also include a communication port **1644** that is configured to transmit and/or receive signals or electrical communication from an external or separate device. The communication port **1644** may be configured to couple to an external device via a cable, adaptor, or other type of electrical connector. In some embodiments, the communication port **1644** may be used to couple the electronic device **1604** with a computing device and/or other appropriate accessories configured to send and/or receive electrical signals. The communication port **1644** may be configured to receive identifying information from an external accessory, which may be used to determine a mounting or support configuration. For example, the communication port **1644** may be used to determine that the electronic device **1604** is coupled to a mounting accessory, such as a particular type of stand or support structure.

Other examples and implementations are within the scope and spirit of the disclosure and appended claims. For example, features implementing functions may also be

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physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Also, as used herein, including in the claims, “or” as used in a list of items prefaced by “at least one of” indicates a disjunctive list such that, for example, a list of “at least one of A, B, or C” means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Further, the term “exemplary” does not mean that the described example is preferred or better than other examples.

The foregoing description, for purposes of explanation, uses specific nomenclature to provide a thorough understanding of the described embodiments. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the described embodiments. Thus, the foregoing descriptions of the specific embodiments described herein are presented for purposes of illustration and description. They are not targeted to be exhaustive or to limit the embodiments to the precise forms disclosed. It will be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. A wearable audio device comprising:
 - an enclosure defining an opening, an interior volume, and an exterior surface;
 - a sealing component coupled to the enclosure and configured to form a sealed passage from the opening to an ear of a user;
 - a speaker positioned within the interior volume and acoustically coupled to the opening;
 - a touch sensor configured to detect:
 - a first gesture input applied to the exterior surface, the first gesture input having a first movement direction along the exterior surface; and
 - a second gesture input applied to the exterior surface, the second gesture input having a second movement direction along the exterior surface, the second movement direction different from the first movement direction; and
 - a processing unit positioned within the interior volume and operatively coupled to the touch sensor and the speaker and configured to:
 - cause the speaker to provide an audio output;
 - cause the speaker to alter the audio output in response to the first gesture input; and
 - cause the speaker to alter the audio output in response to the second gesture input.
2. The wearable audio device of claim 1, wherein:
 - altering the audio output in response to the first gesture input comprises increasing a volume of the audio output; and
 - altering the audio output in response to the second gesture input comprises decreasing the volume of the audio output.
3. The wearable audio device of claim 1, wherein the first movement direction is opposite the second movement direction.
4. The wearable audio device of claim 1, wherein:
 - the enclosure comprises:
 - a body; and
 - a stem extending from the body and defining at least a portion of the exterior surface;
 - the speaker is positioned at least partially in the body; and
 - the touch sensor is configured to detect gesture inputs applied to the stem.

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5. The wearable audio device of claim 4, wherein each of the first gesture input and the second gesture input are applied to the stem.

6. The wearable audio device of claim 1, wherein:

- the wearable audio device further comprises a microphone positioned in the enclosure and configured to detect an audio input; and
- the processing unit is further configured to alter the audio output in response to the audio input.

7. The wearable audio device of claim 6, wherein:

- the enclosure comprises:
 - a body; and
 - a stem extending from the body; and
- the microphone is positioned in the stem.

8. A wearable audio device comprising:

- an enclosure defining an exterior surface and a touch-sensitive region along the exterior surface;
- a speaker at least partially within the enclosure and configured to produce an audio output;
- a compliant member coupled to the enclosure and configured to retain the wearable audio device in a user's ear;
- a sensing system configured to detect:
 - a first gesture input applied within the touch-sensitive region and having a first motion direction; and
 - a second gesture input applied within the touch-sensitive region and having a second motion direction, different from the first motion direction; and
- a processing unit at least partially within the enclosure and configured to alter the audio output based at least in part on detection of at least one of the first gesture input or the second gesture input.

9. The wearable audio device of claim 8, wherein the enclosure is configured such that the exterior surface is positioned outside of the user's ear when the wearable audio device is being worn.

10. The wearable audio device of claim 8, wherein the processing unit is configured to:

- alter the audio output in a first manner in response to the first gesture input; and
- alter the audio output in a second manner, different from the first manner, in response to the second gesture input.

11. The wearable audio device of claim 10, wherein:

- altering the audio output in the first manner comprises increasing a volume; and
- altering the audio output in the second manner comprises decreasing the volume.

12. The wearable audio device of claim 8, wherein:

- the compliant member is coupled to a first side of the enclosure; and
- the touch-sensitive region is positioned along a second side of the enclosure opposite the first side.

13. The wearable audio device of claim 8, wherein:

- the enclosure comprises:
 - a body; and
 - a stem extending from the body; and
- the wearable audio device further comprises a microphone positioned in the stem and configured to receive an audio input.

14. The wearable audio device of claim 13, wherein the processing unit is further configured to alter the audio output in response to the audio input.

15. A wearable device comprising:

- an enclosure;
- a speaker within the enclosure;

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an earpiece portion extending from the enclosure and defining an acoustic passage for audio output from the speaker;

a touch sensing system at least partially within the enclosure and defining a touch sensitive region along an exterior surface of the enclosure, the touch sensing system configured to detect:

- a first gesture within the touch sensitive region and having a first direction;
- a second gesture within the touch sensitive region and having second direction, different from the first direction; and
- a processing unit at least partially within the enclosure and configured to:
 - alter the audio output in a first manner in response to detecting the first gesture; and
 - alter the audio output in a second manner, different from the first manner, in response to detecting the second gesture.

16. The wearable device of claim 15, wherein the touch sensitive region has a first surface texture that is distinct from a second surface texture of a surrounding surface of the enclosure.

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17. The wearable device of claim 15, wherein altering the audio output in the first manner comprises increasing a volume of the audio output, and altering the audio output in the second manner comprises decreasing the volume of the audio output.

18. The wearable device of claim 15, wherein:

the wearable device further comprises a microphone at least partially within the enclosure and configured to receive an audio input; and

the processing unit is further configured to alter the audio output in response to the audio input.

19. The wearable device of claim 18, wherein:

the microphone is a first microphone; and

the wearable device further comprises a second microphone at least partially within the enclosure and configured to detect ambient noise.

20. The wearable device of claim 19, wherein the processing unit is further configured to acoustically cancel the ambient noise from the audio output.

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